Atmosic Production Tool

User Guide

SUMMARY: This document describes the mass production environment to test ATM2/ATM3 devices on production lines without RF testing equipment.



Atmosic™

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1. Overview

This document describes the mass production environment to test ATM2/ATM3 devices on production lines without RF testing equipment. It consists of both hardware and software items:

- Atmosic Mass Production Tool (AtmMpTool.exe), an application software on MS Windows.
- Atmosic Production Tester (APT), provides the capabilities of Bluetooth LE testing with DUT controlled by AtmMpTool.exe.
- Atmosic Download Board (DL), provides the capabilities of DUT HCI control and DUT firmware, NVDS and OTP programming by AtmMpTool.exe.

The Atmosic Mass Production Tool and ATP provide these functionalities:

- RF Tx Test
- RF Rx Test
- RF Frequency Test
- Crystal Trim
- Write firmware and NVDS

2. Environment Setup for the APT

A complete set of the testing suite includes:

- PC x1 (Windows 10)
- Atmosic Production Tester (APT) (provided by Atmosic)
- Dipole antenna[1] x 1
- Type-B USB cable x 1
- 2x10 Test interface cable x 1
- MPTool software

Figure 2-1 depicts the APT.

[1] The dipole antenna should be connected to SMA_DIR by default.

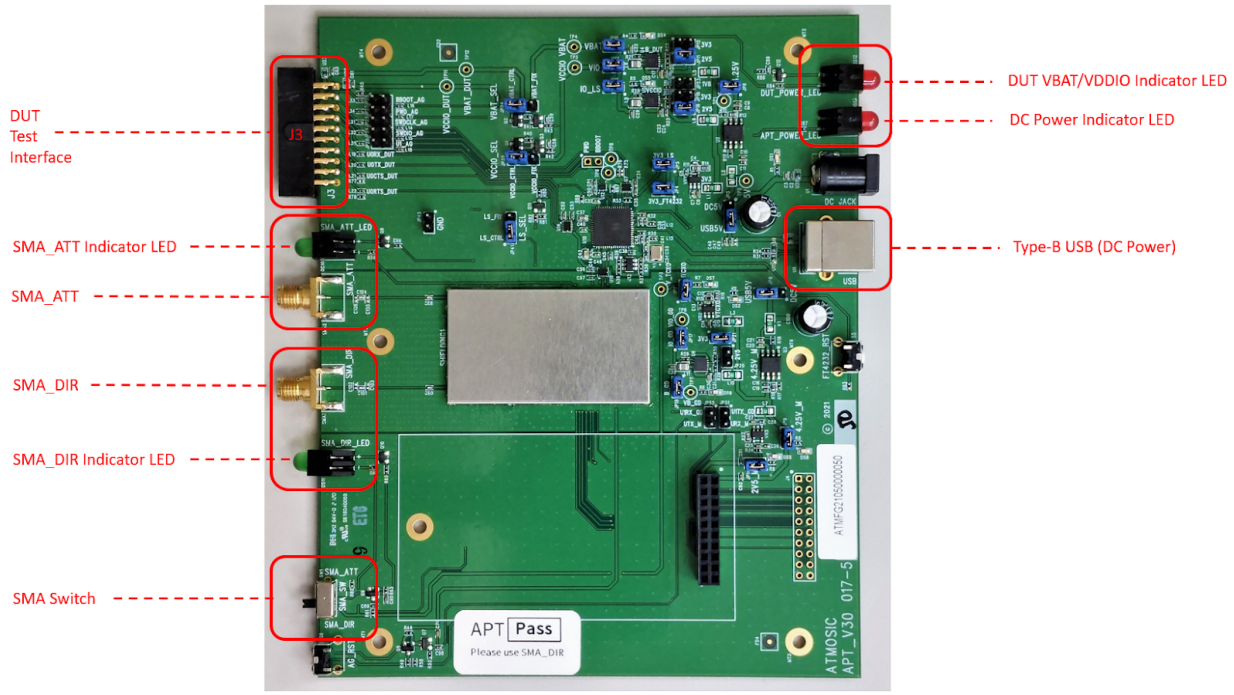


Figure 2-1 Atmosic Production Tester Board Configuration

Figure 2-2 shows a connection setup between PC, APT and DUT:

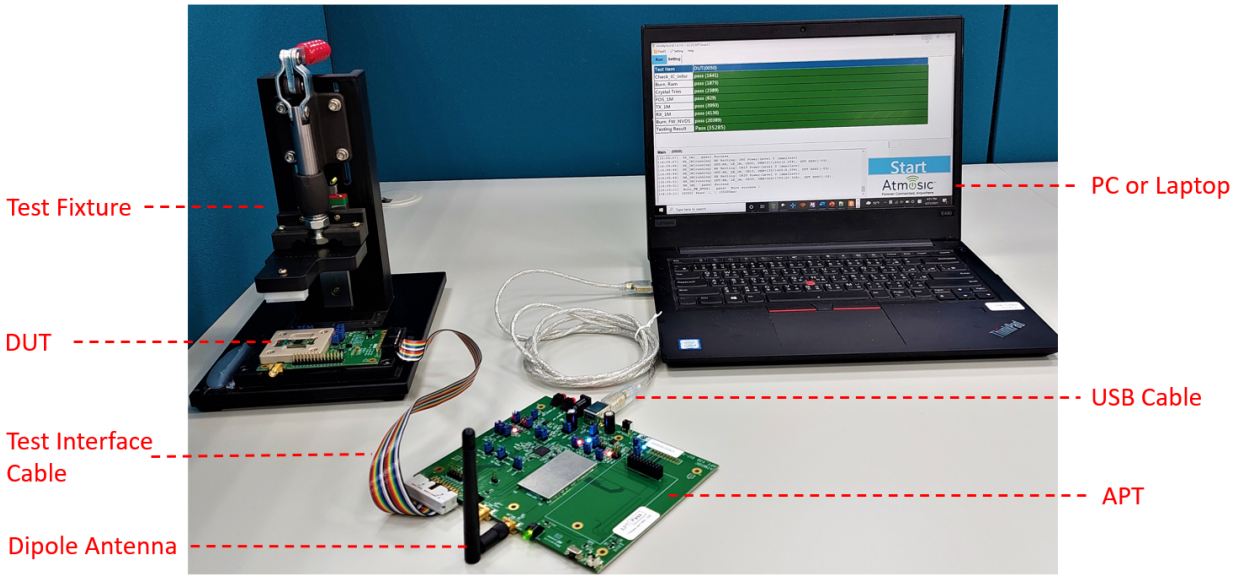


Figure 2-2 Connection Setup Between PC, APT and DUT

3. Environment Setup for DL (Download Board)

A complete set of the testing suite includes:

- PC x1 (Windows 10)
- Atmosic Mass Production Download Board (DL) (provided by Atmosic)
- Type-B USB cable x 1
- 2x10 Test interface cable (connect from DUT test pin to J1 of APT) x 1
- MPTool software

The DL_V4 picture is shown in Figure 3-1.

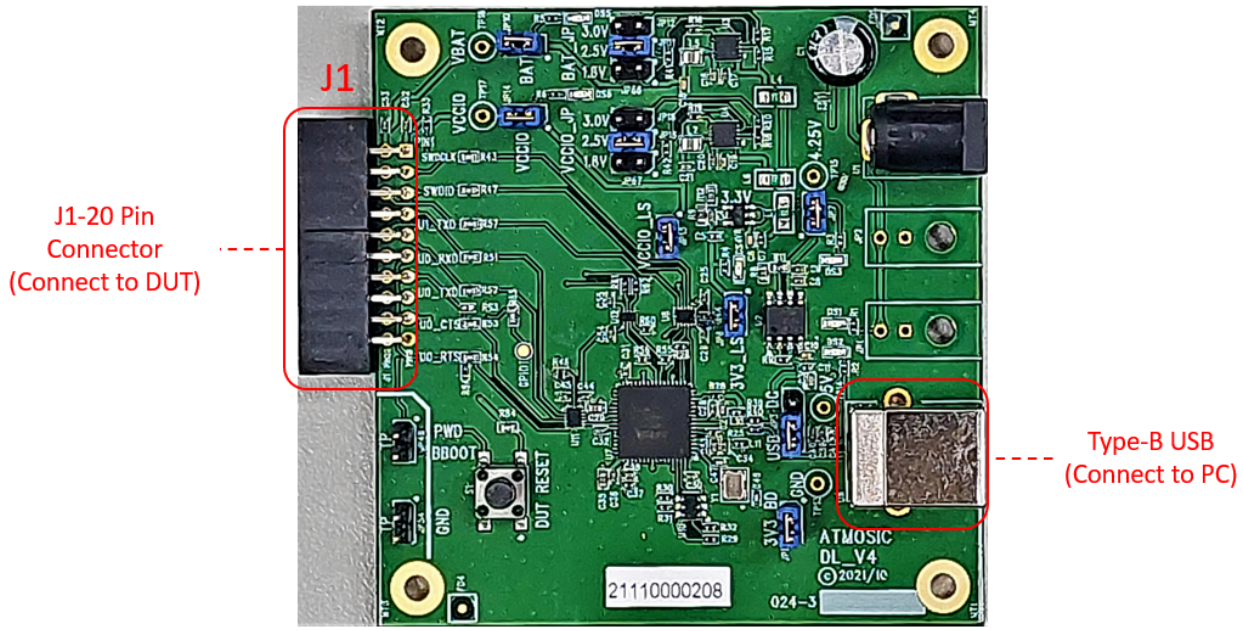


Figure 3-1 Mass Production Download Board (DL) Configuration

Figure 3-2 shows a connection setup between PC, DL and DUT.



Figure 3-2 Connection Setup Between PC, DL and DUT

3.1 DL Board Versions

3.1.1 DL_V4

DL_V4 is currently supported by MP Tool versions up to V2.0.11. The default jumper settings for DL_V4 are shown in Figure 3-3.

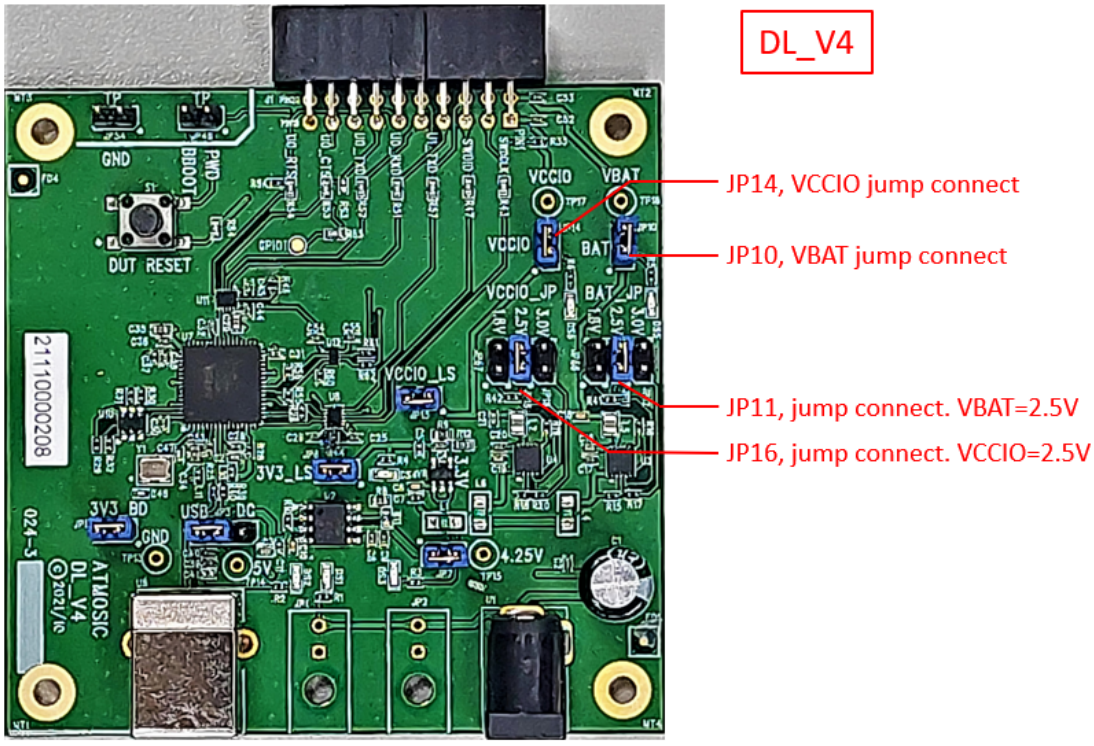


Figure 3-3 DL_V4 Critical Jumper Setting

3.1.2 DL_V6a

DL_V6a is backward compatible with DL_V4. Both DL_V4 and DL_V6a do not support the power control function available in DL_V6. The default jumper settings for DL_V6a are shown in Figure 3-4.

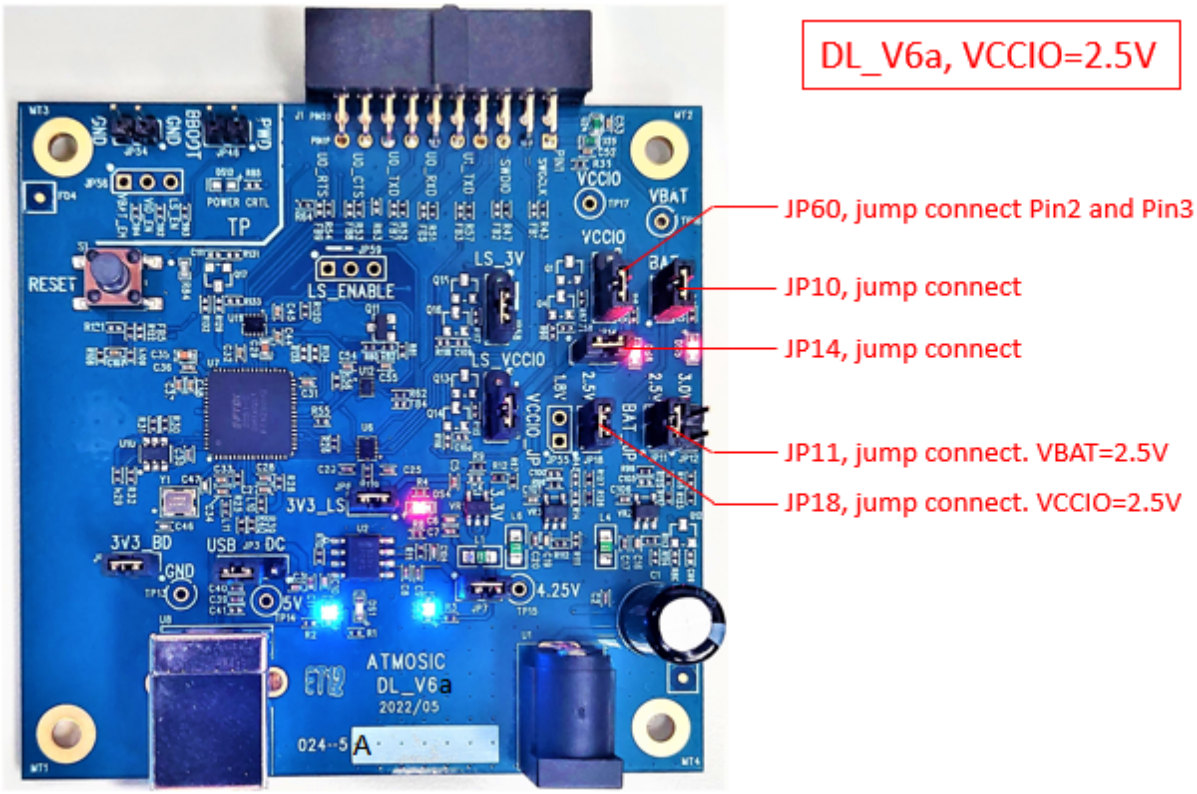


Figure 3-4 DL_V6a Critical Jumper Setting

3.1.3 DL_V6

DL_V6 adds a power control function to ensure that the DUT is powered off before the MPtool test program runs and DL_V6 supports either 2.5 V or 1.8 V I/O voltages. To use DL_V6, the MP Tool version must be V2.0.12 or higher.

Since the I/O voltage written to OTP requires a 2.5 V I/O voltage level, the default jumper settings are shown in Figure 3-5, which support a 2.5 V I/O voltage.

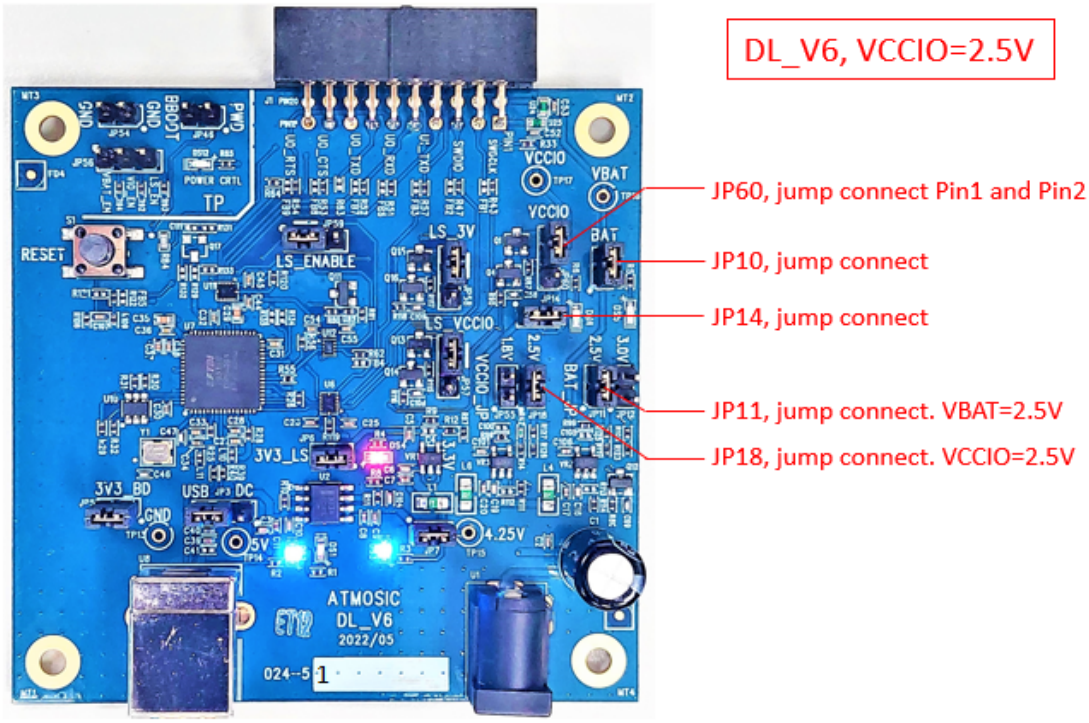


Figure 3-5 DL_V6 with 2.5 V I/O Voltage Critical Jumper Setting

If the I/O voltage needs to use 1.8 V I/O, the recommended jumper settings diagram is shown in Figure 3-6, which support 1.8 V I/O.

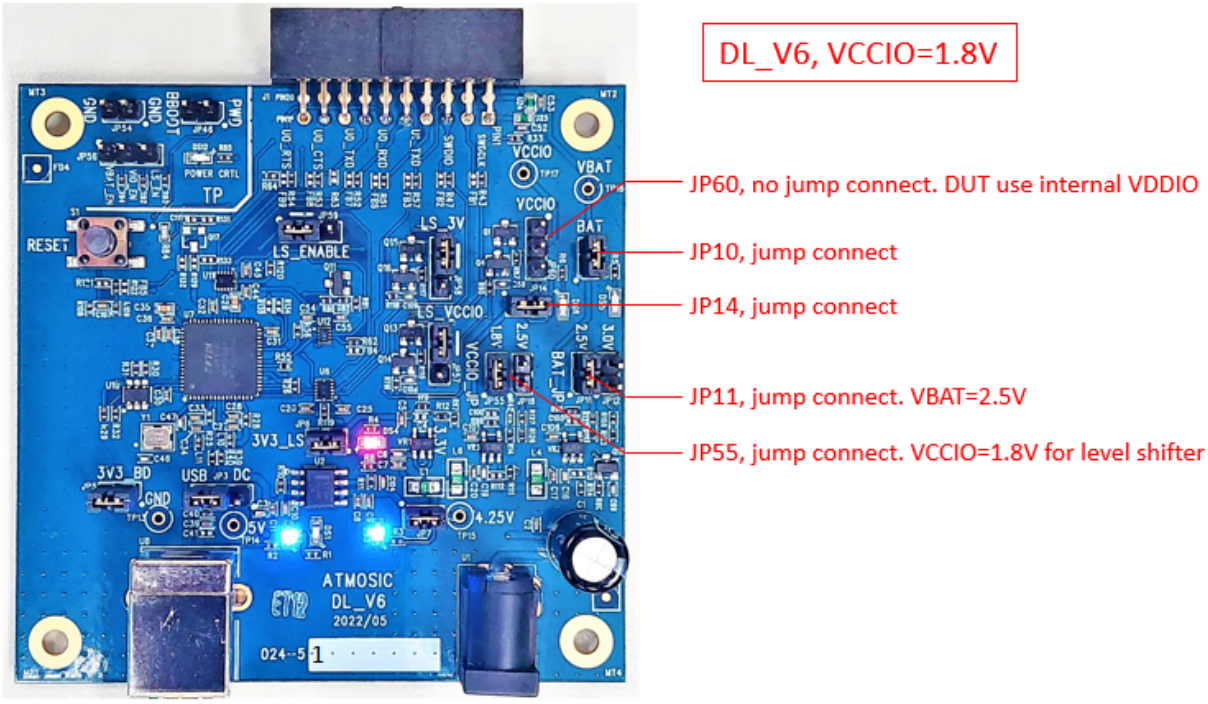


Figure 3-6 DL_V6 with 1.8 V I/O Voltage Critical Jumper Setting

4. Hardware Setup

There is a Type-B USB port that is connected to a FTDI FT4232H, which provides four UART ports.

Test point connector is the interface between the DUT and APT/DL and as shown in Figure 4-1. It provides a host computer access to some commonly needed interfaces on an Atmosic chip based DUT, such as

- Power and Ground:
  - VBAT
  - VCCIO
  - GND
- Serial Wire Debug (SWD):
  - SWDCLK
  - SWDIO
- 4-wire HCI UART for RF Test control
  - UART0_TXD
  - UART0_RXD
  - UART0_CTS
  - UART0_RTS
- Status Control signal:
  - PWD: Used to Reset/PWD the DUT
  - BBOOT: Benign Boot, pull high to keep the DUT MCU in an idle state during boot.
  - TMC: Test Mode Control, pull high to enable Test Mode, which enables SWD. If TMC is already pulled high on the DUT side, this pin doesn't need to be connected.
- Debug UART:
  - UART1_TXD, optional, for debugging only.

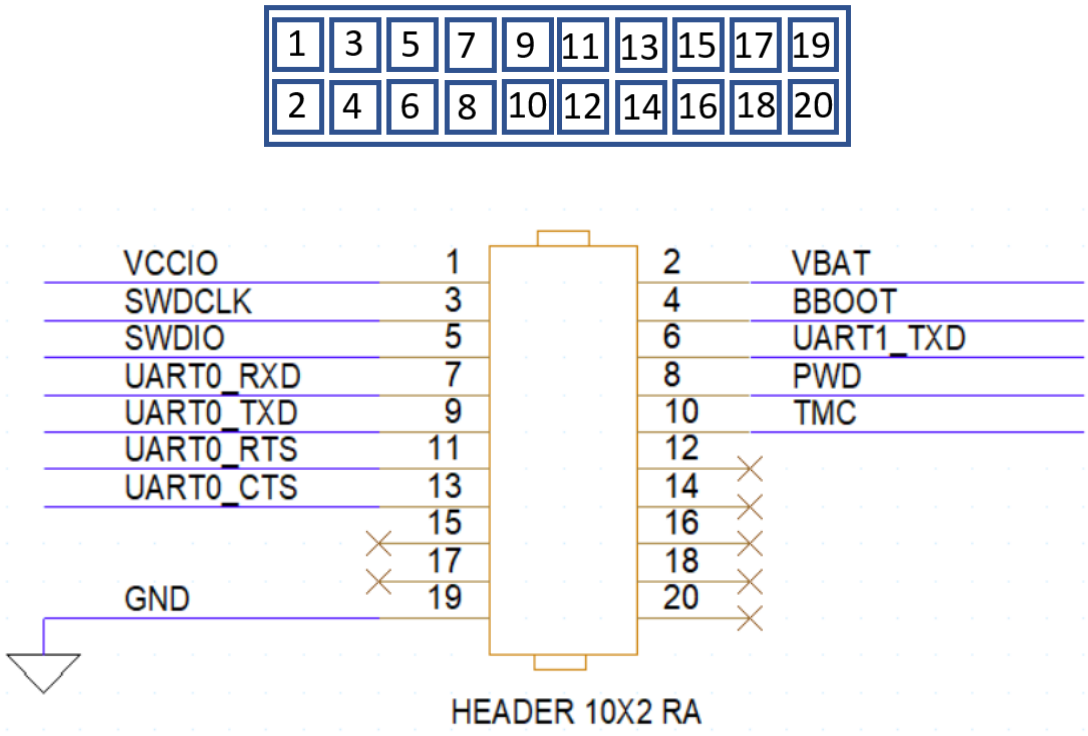


Figure 4-1 Test Point Connector Pin Definition

5. Software Setup

5.1 Uninstall any previous versions of the MPTool

If any earlier version of the MP Tool was installed, execute its uninstaller, which should be at C:\AtmosicMP\unInst_MP.exe and wait for the uninstaller to complete, then click Close. See Figure 5-1.

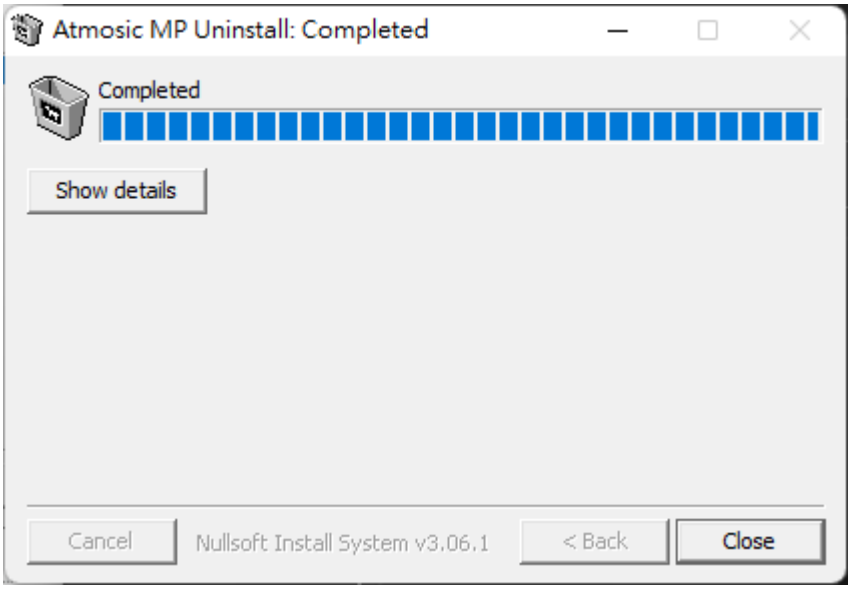


Figure 5-1 Uninstall Previous MP Tool Version Completed

5.2 Install Atmosic Production Tool

The setup package of Windows Installer For Atmosic Mass Production (MP) Tool is a compressed file named **AtmosicMP_Inst_20xxxxxx_vx.x.x.x.zip**. Please extract the file and double click the “AtmosicMP_Inst.exe” to start the Atmosic MP Tool Installer.

The Atmosic MP Tool will occupy 50 MB of disk space and should be located in the “C:\AtmosicMP” folder. Please ensure enough disk space is available before installing the Atmosic MP Tool. Subsequently, click the “Install” button to continue. Figure 5-2 shows the Atmosic MP Tool Destination Folder.

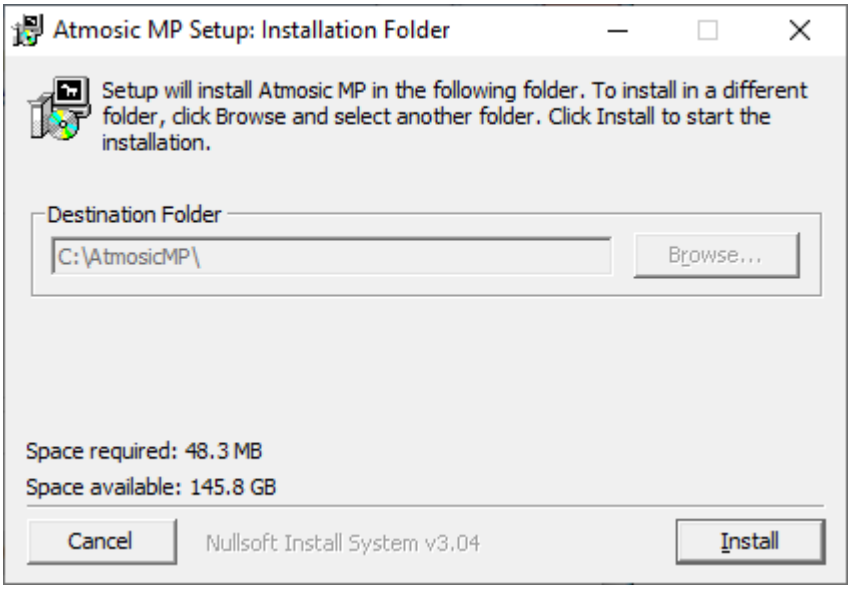


Figure 5-2 Atmosic MP Tool Destination Folder

Figure 5-3 shows the screen after the installer is completed. The installer will create a folder named AtmosicMP in “Start Menu\Atmosic”, and create shortcuts to invoke Atmosic MP Tool features that will be discussed in the next section.

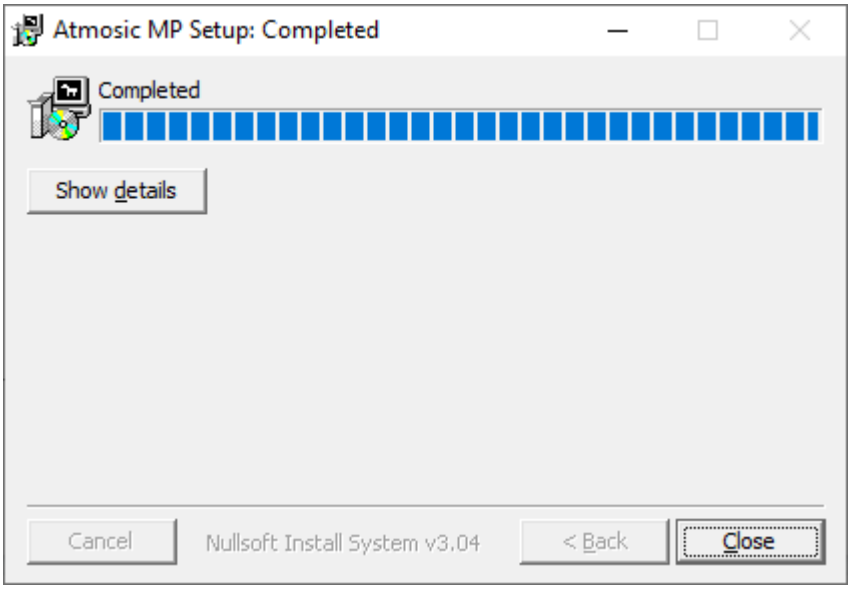


Figure 5-3 Atmosic MP Tool Setup Completed

5.3 Atmosic Production Tool Shortcuts

The Atmosic Production Tool Installer will create shortcuts in “Start Menu\AtmosicMP” after the installer is completed, and the shortcuts include install or remove MFx interface, and MP API document as shown in Figure 5-4.

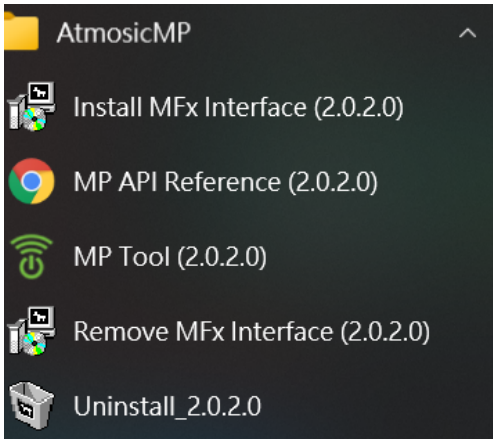


Figure 5-4 Start Menu Shortcuts

5.4 Install MFx Interface

Please connect the Atmosic APT (Atmosic Production Tester) to the laptop and wait a while (~ 30 seconds) to install the driver, which usually exists in the Windows default. If your computer cannot install this automatically, please download it from FTDI website and install it yourself.

Next, open the Windows Device Manager (open the “Run” dialog box by pressing and holding the Windows key, then press the R key.

Next, type in “devmgmt.msc” then click the OK button to verify whether the Atmosic MFx interface existed or not.

If there are 4 COM ports listed in the Device Manager after plug-in of the Atmosic APT to laptop, as shown in Figure 5-5, please click the “Install MFx Interface " shortcut in “Start Menu/Atmosic” folder. The COM port numbers were assigned by the Windows system.

If not, please download the FTDI driver and install it before installing the MFx interface driver.

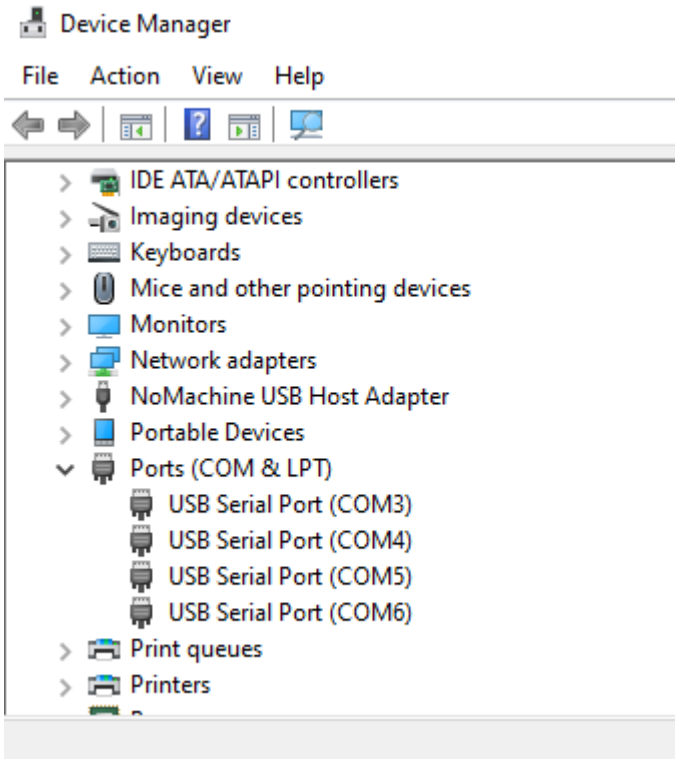


Figure 5-5 COM Port Setup in Device Manager

After installing the Atmosic MFx driver, the first COM port (such as COM3 in Figure 5-6) would be assigned to Atmosic MFG (Interface 0) for programming firmware, NVDS or OTP.

The second COM port (such as COM4 in Figure 5-6) is used to dump logs from DUT.

The 3rd COM port (such as COM5 in Figure 5-6) is used to communicate between Atmosic DUT and laptop through HCI commands.

The 4th COM port (such as COM6 in Figure Figure 5-6) is used to communicate between Atmosic APT and laptop through HCI commands.

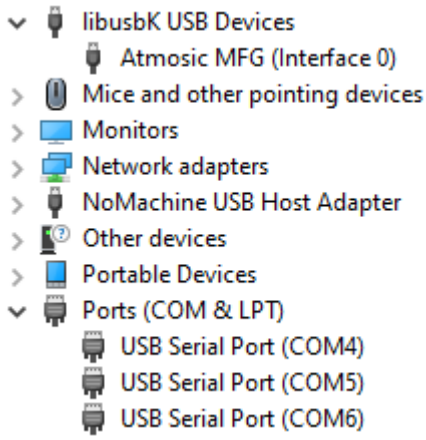


Figure 5-6 COM Port Assignment

5.5 Uninstall MFx Interface

All Atmosic MFx Interfaces that are installed in the Windows system will be uninstalled after performing the Uninstall MFx Interface program.

6 Adjust Serial Port Latency Timer

Setting a lower latency timer will speed up the testing process when the test item uses the UART HCI command.

First, open the Windows Device Manager (open the “Run” dialog box by pressing and holding the Windows key, then press the R key.

Next, type-in “devmgmt.msc” then click the OK button to verify whether the Atmosic MFx interface existed or not.

As shown in Figure 6-1, for every “USB Serial Port” , please right click and select “Properties”.

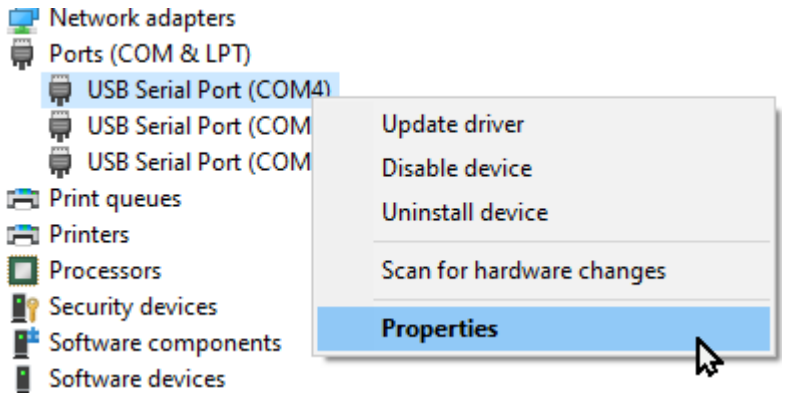


Figure 6-1 COM Port Setup in Device Manager

See Figure 6-2, Click “Advanced…”

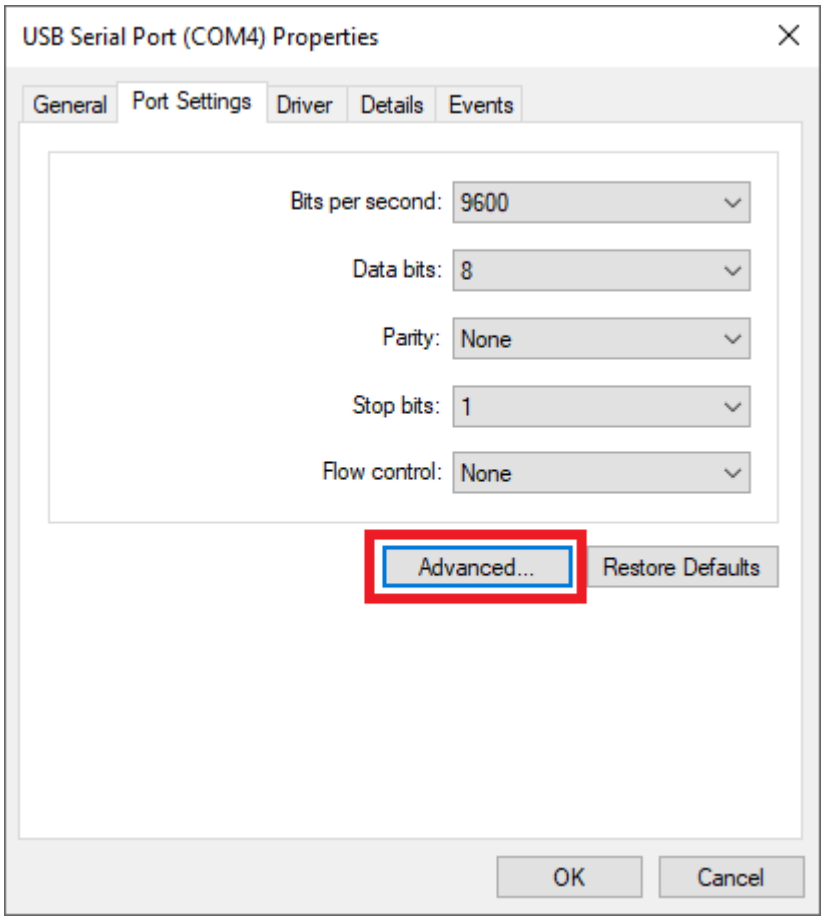


Figure 6-2 USB Serial Port Properties

See Figure 6-3, Change “Latency Timer” to “1 msec” (millisecond)

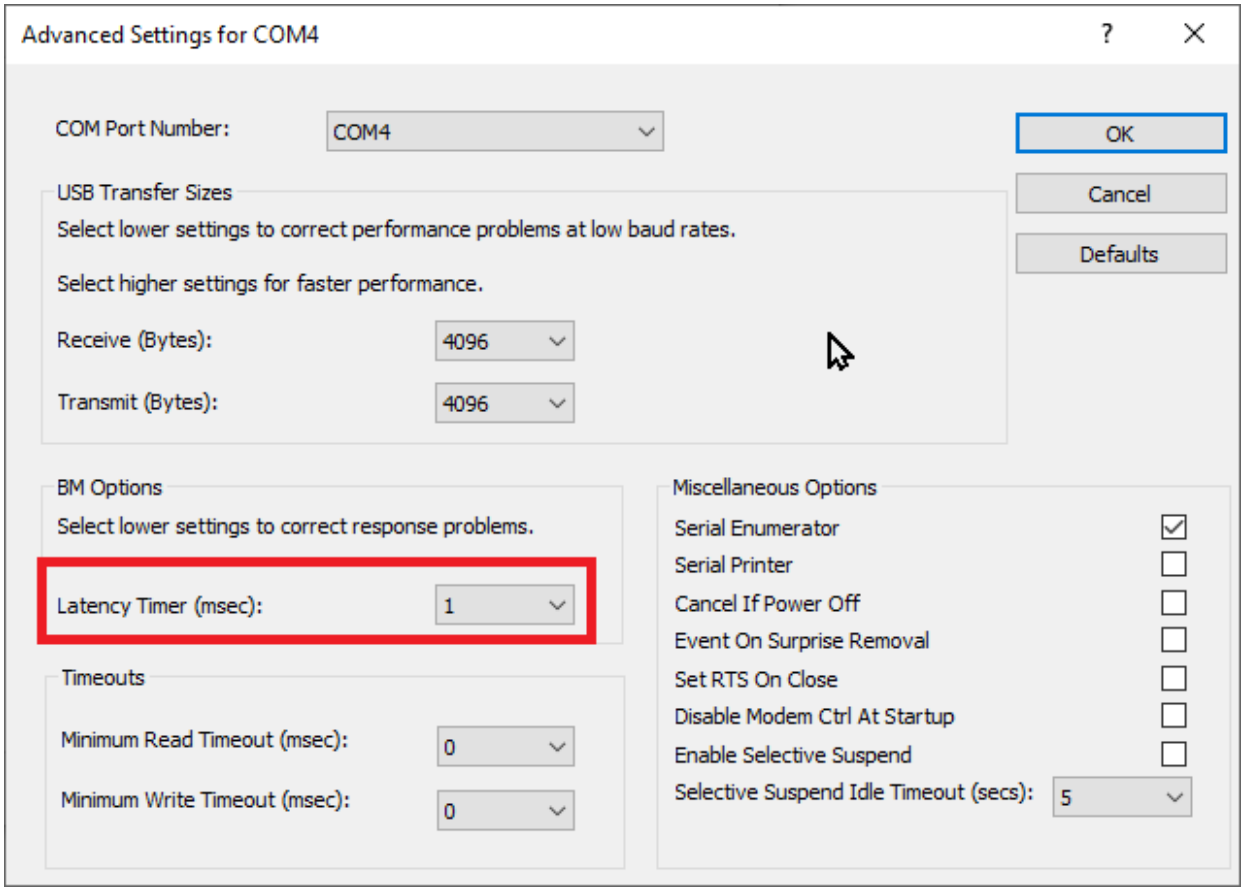


Figure 6-3 Advanced Setting for COM ports

7. GUI Mode Quick Start

In GUI mode, please refer to the setting in this chapter first. Following sections are steps to quickly start testing.

7.1 Select ATM file

If you want to program flash, please choose an ATM file to the option as shown in Figure 7-1.

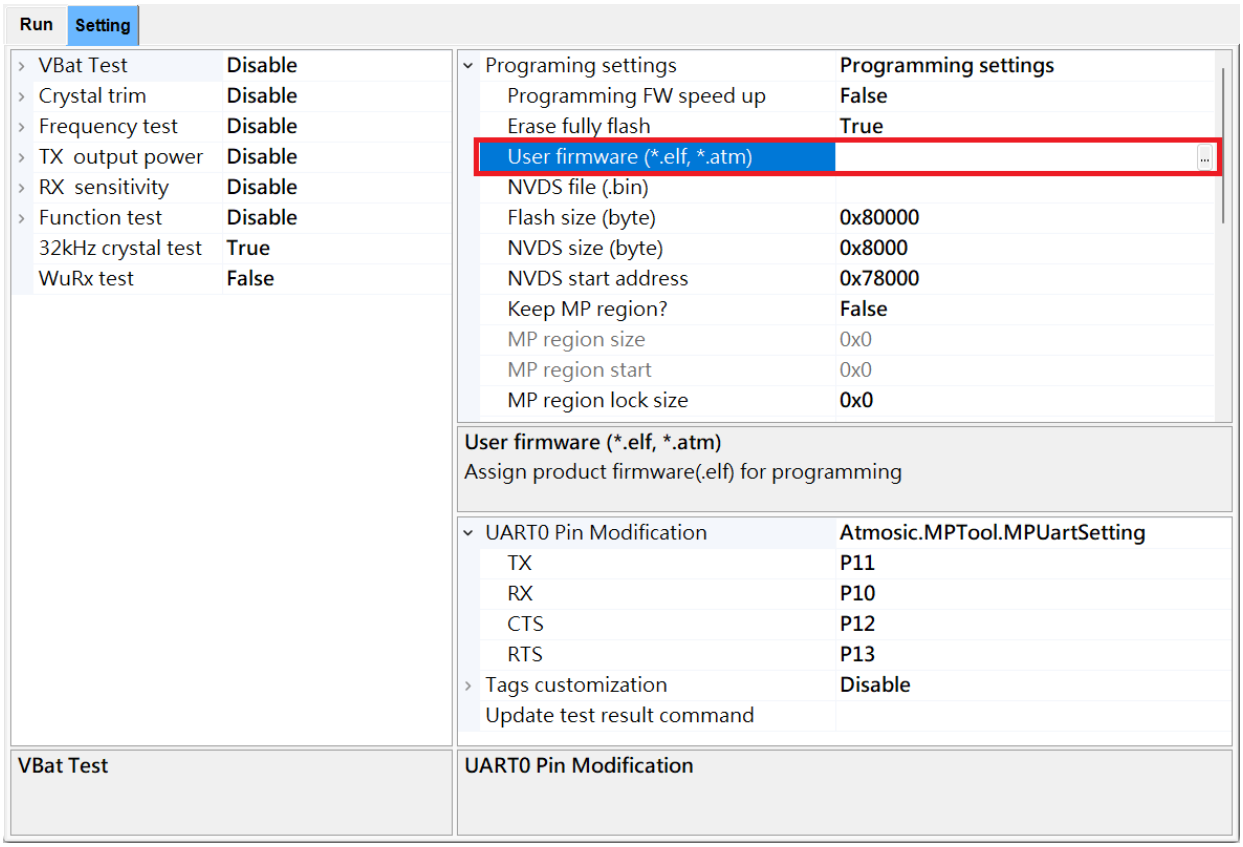


Figure 7-1 Select ATM File

When you select an ATM file, if a pop-up message box “This atm file has NVDS, do you want to load this NVDS” means this .atmlfile has NVDS data, you can click “Yes” to export bin file and auto load NVDS file to the option “NVDS file(.bin)”.

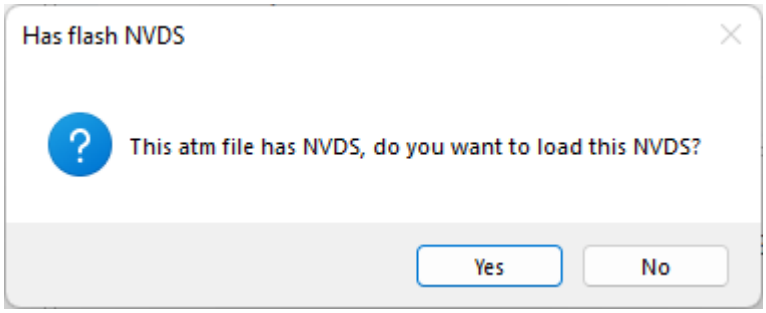


Figure 7-2 Load NVDS

7.2 RF testing with APT

7.2.1 Choose testing options

There are 4 test items: “Crystal trim”, “Frequency test", “TX output power” and “RX sensitivity”, you can change these items to “Enable” to enable these items.

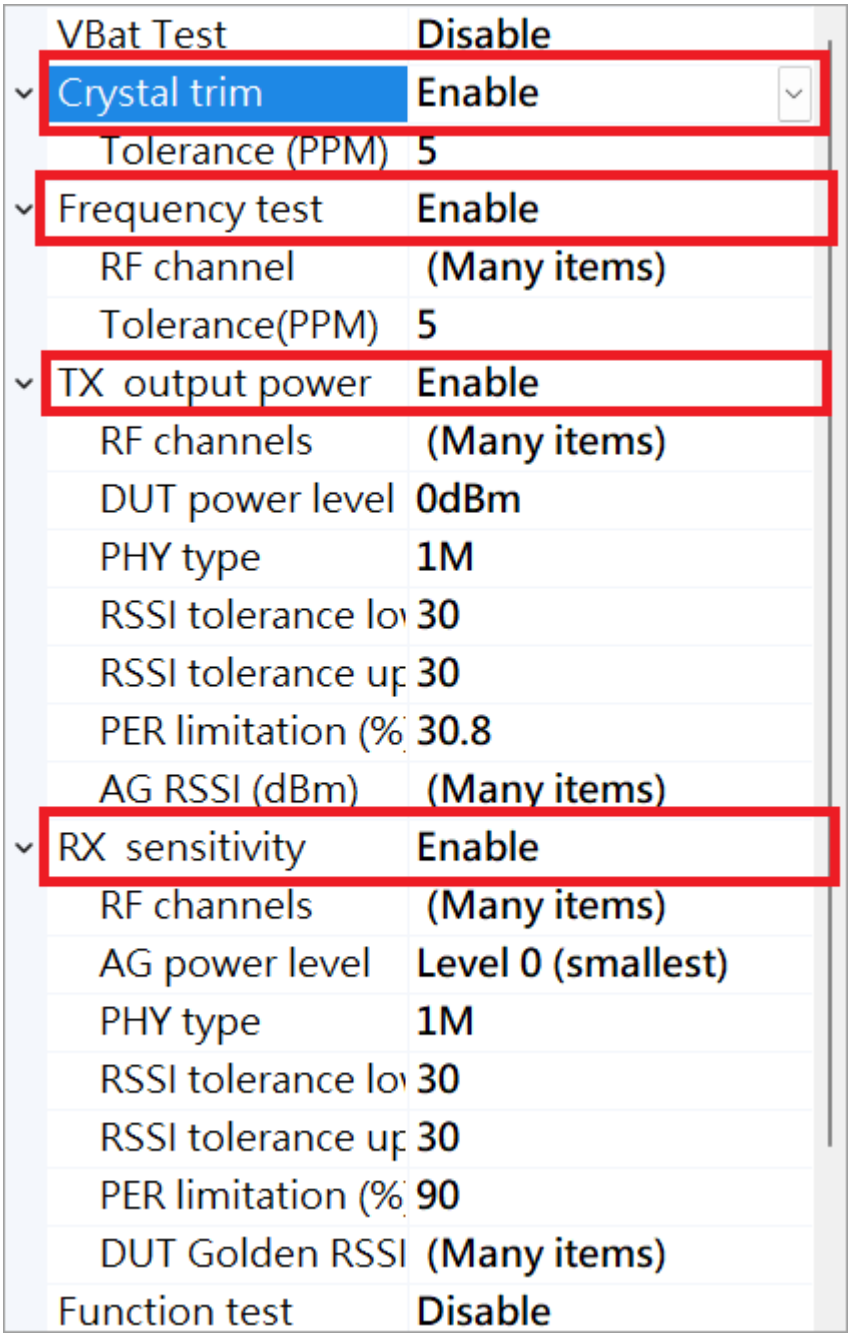


Figure 7-3 Choose Test Options

When you select any RF test, it will trigger the “Burn_Ram” process, and it will use the HCI command via UART to perform RF testing. You can see “Burn_Ram” process shows on “Run” page.

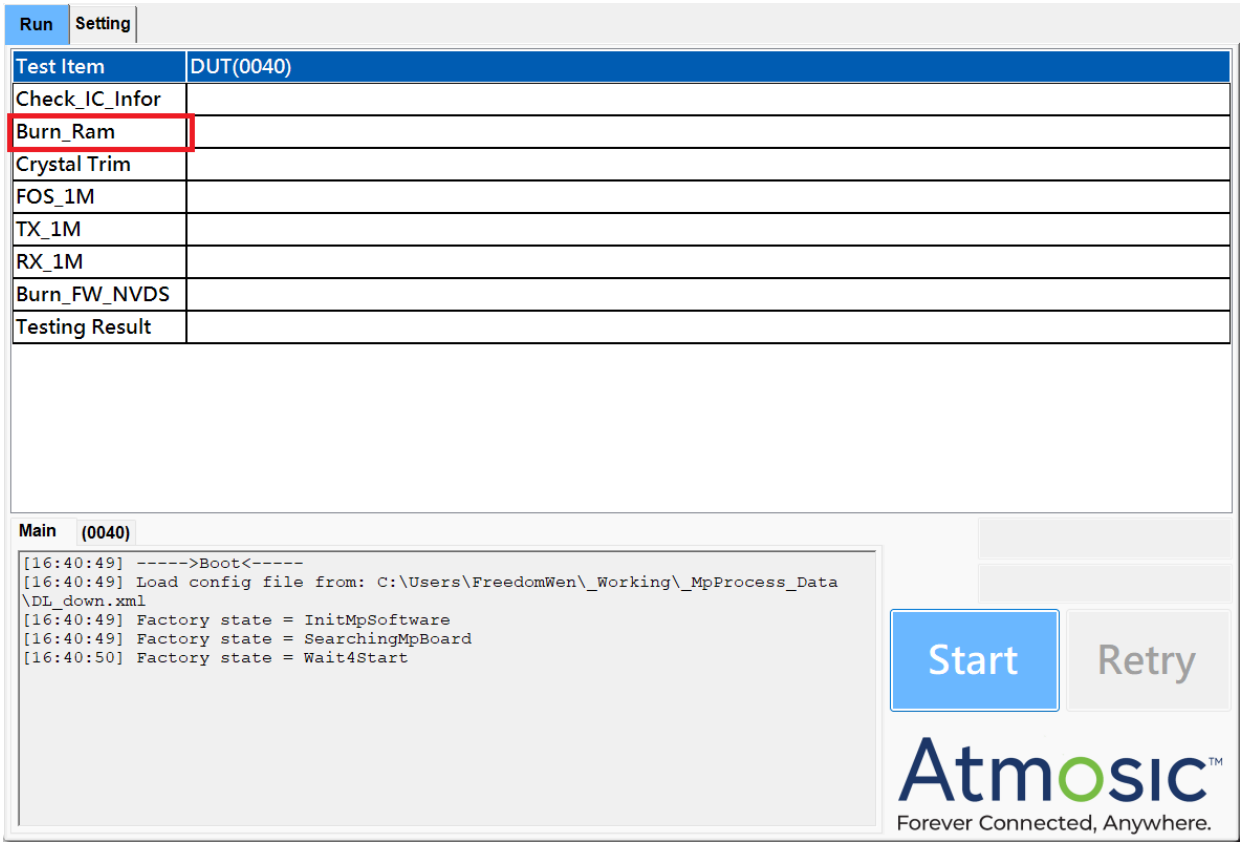


Figure 7-4 Burn_RAM Process

7.2.2 Choose testing options

When you choose RF testing, you need to use UART to send HCI commands, there please set UART pins first.

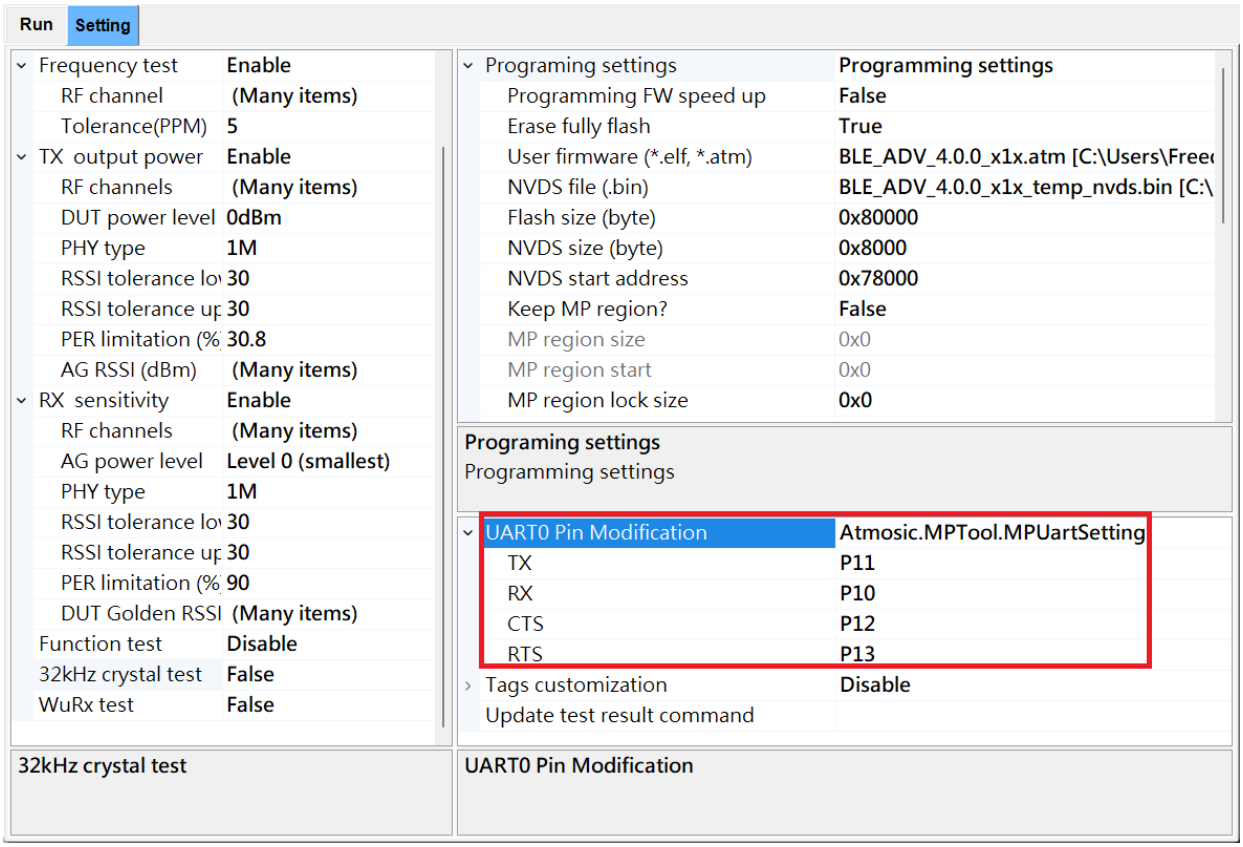


Figure 7-5 Select ATM File

For more suggestions on UART pin setting, please refer to the UART0 Pin Modification section.

8. Atmosic Mass Production Tool User Interface

The tool is used for mass production tests on ATM2/ATM3 devices. The procedure includes crystal trim, frequency test, TX test, RX test and program firmware.

After installation, users can launch AtmMpTool.exe from the Start menu of windows.

AtmMpTool will generate production log files for each test. These log files are saved in the log folder. It can be opened by clicking the “Production Log Directory” menu item.

See Figure 8-1.

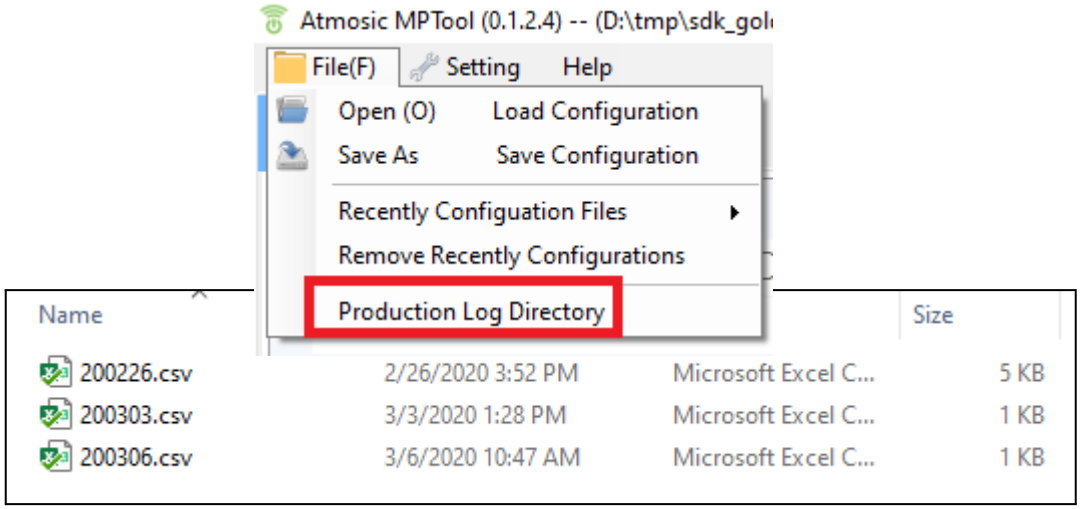


Figure 8-1 Atmosic Mass Product Tool User Interface

The production logs were written in .csv format and one file every day.

The GUI contains three parts:

- [Menus] : used for saving config, refreshing devices, unlocking Setting Tab, and selecting language.
- [Run Tab] : used for controlling the running of tests.
- [Setting Tab] : used for setting the parameters of test items.

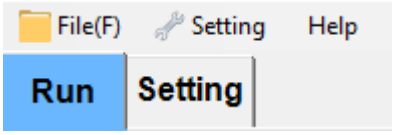


Figure 8-2 MP Tool Run Menu

Quick Start Operation Steps:

1) Please unlock Setting Tab (password is required and initial value is “Atmosic123”)
2) Set the test parameters of enabled test items in Setting Tab (Or load setting from File)
3) Click the “START” button on the Running Tab to start the test on DUTs.

8.1 Menus

8.1.1 File Menu

Here are five menu items as shown in Figure 8-3 and describe as below:

- [Open] : Load configuration file into application.
- [Save as] : Save current test configuration to file.
- [Recent Configuration Files] : Recent list of configuration files.
- [Remove Recent Configurations] : Remove the recent list.
- [Open Production Log Folder] : Open production logs' folder.

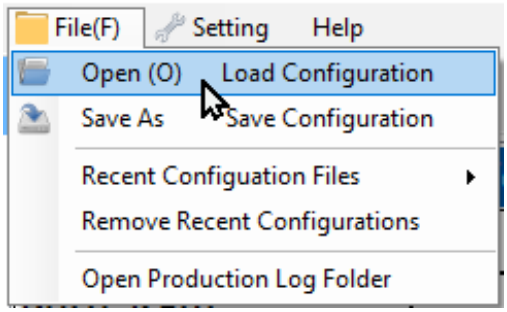


Figure 8-3 MP Tool File Menu

8.1.2 Setting Menu

Here are four menu items as shown in Figure 8-4 and describe as below:

- [Unlock]: Unlock the Setting Tab from read only mode.
- [Stop]: Stop testing after testing is started.
- [Refresh APT/DL]: Re-search the APT. Generally, APTs are searched automatically during program startup. This is used when a user plugs in/out APTs without relaunching AtmMptool.exe.
- [Command mode]: Enter command line mode, user can perform the unitest by typing predefined commands.

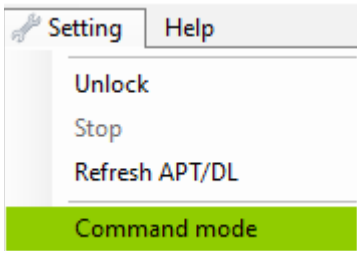


Figure 8-4 MP Tool Setting

8.1.3 Help Menu

Figure 8-5 shows the MP Tool Help Menu.

- [Language]: Switch the language between Simplified Chinese and English.
- [About Atmosic Manufacturing Tool] : About tool information.

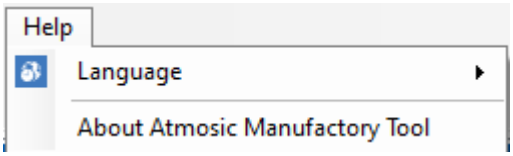


Figure 8-5 MP Tool Help Menu

8.2 Run Tab

There are three main parts in Run Tab as shown in Figure 8-6:

1) Start/Stop button, used for controlling each iteration button.
2) Debug log window, show detail log of the current test.
3) Test progressing table, indicate the current testing schedule.

Run Tab before testing:

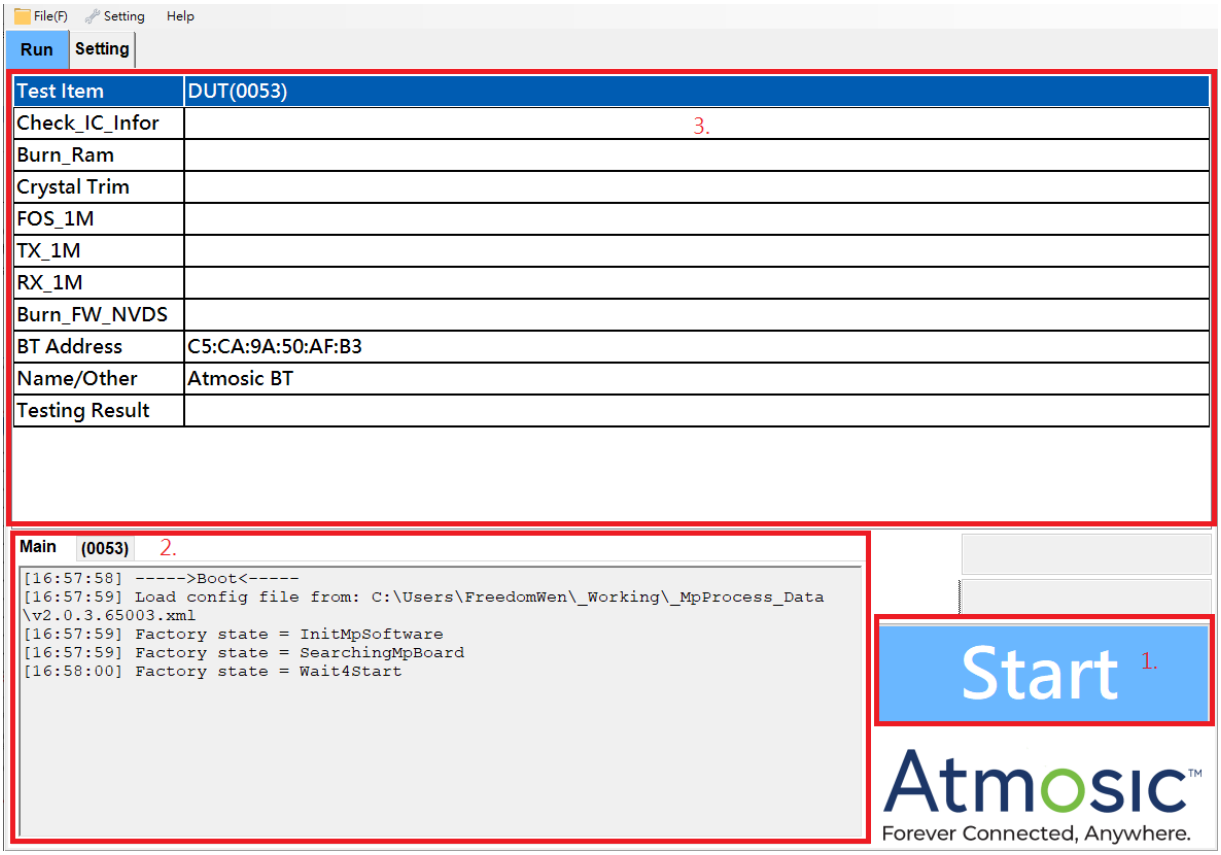


Figure 8-6 Run Tab Before Testing

Before running, all names of test items will show in the first column of the progressing table for the user to check.

Run Tab under testing:

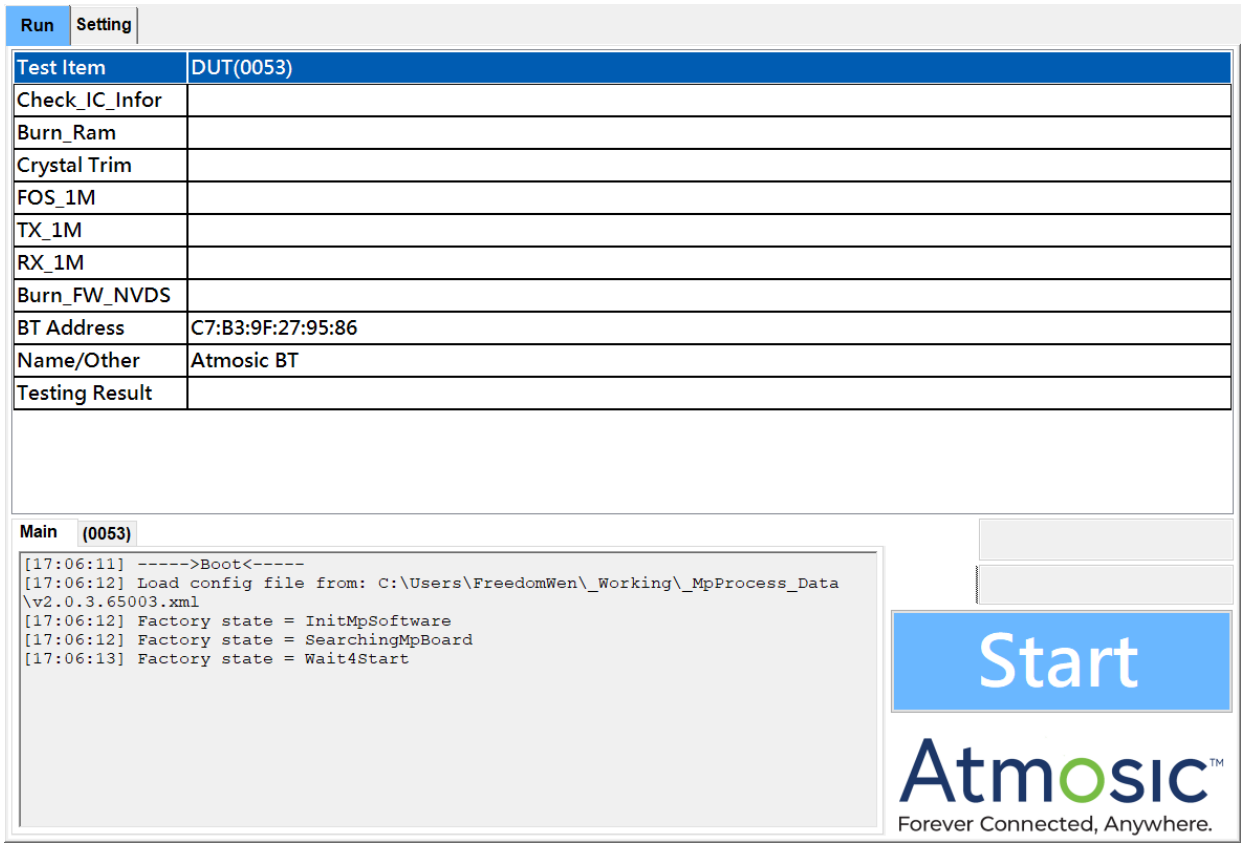


Figure 8-7 Run Tab Under Testing

When running, the corresponding row will show pass or fail and its execution time in milliseconds. See Figure 8-7.

Run Tab after testing success:

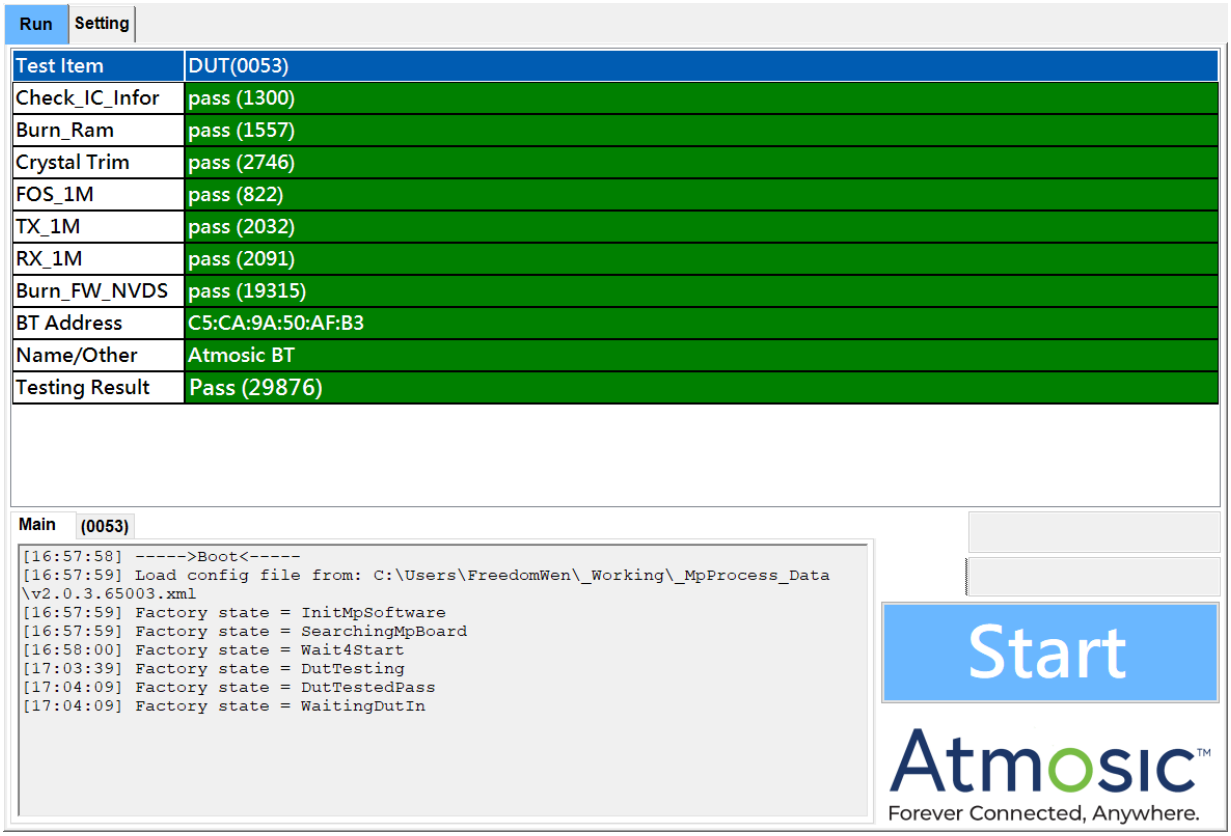


Figure 8-8 Run Tab After Testing Success

After it is finished and all passes, the background will become green. See Figure 8-8.

Running tab after testing fail:

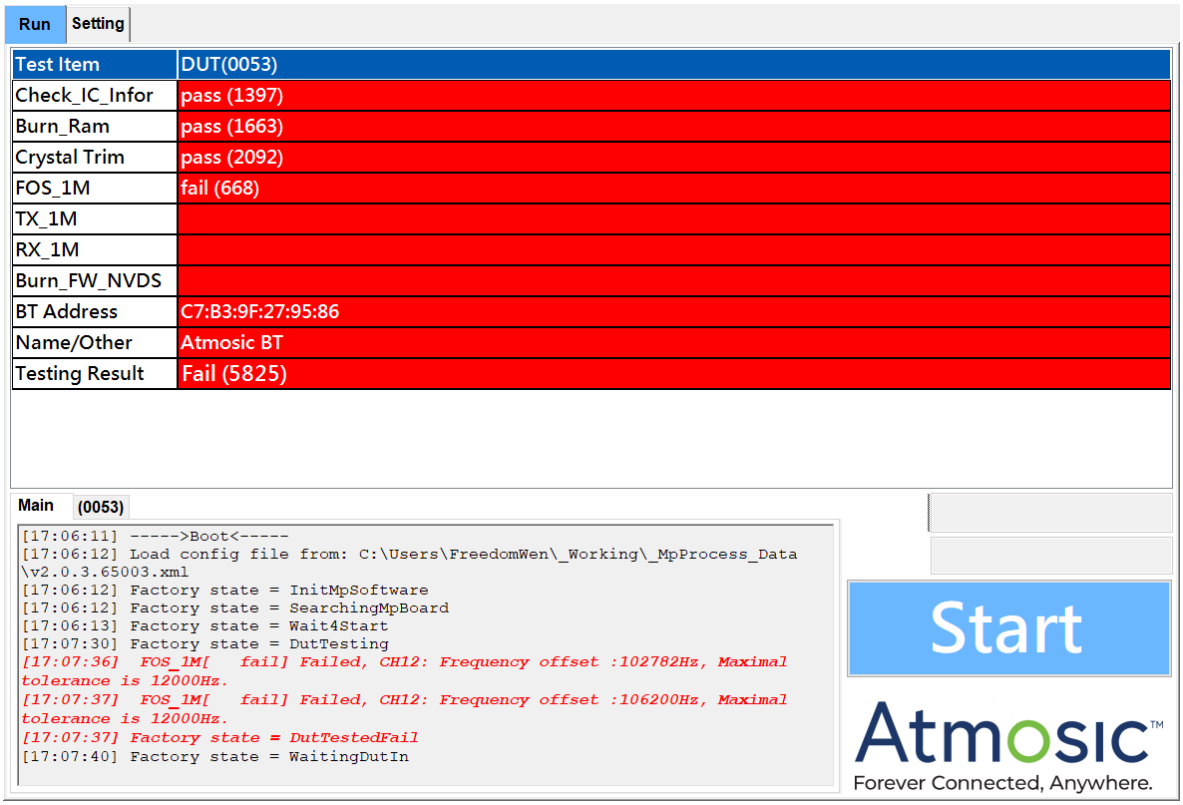


Figure 8-9 Run Tab After Testing Fail

After it is finished and some items fail, the background color will become red. See Figure 8-9.

8.3 Setting Tab

There are three main parts in Setting Tab as shown in Figure 8-10:

1) Setting for Testing.
2) Setting for Programming flash, OTP and MP region.
3) Setting for BT address and name.

To start setting the parameter, unlock the Tab by clicking on the unlock menu item.

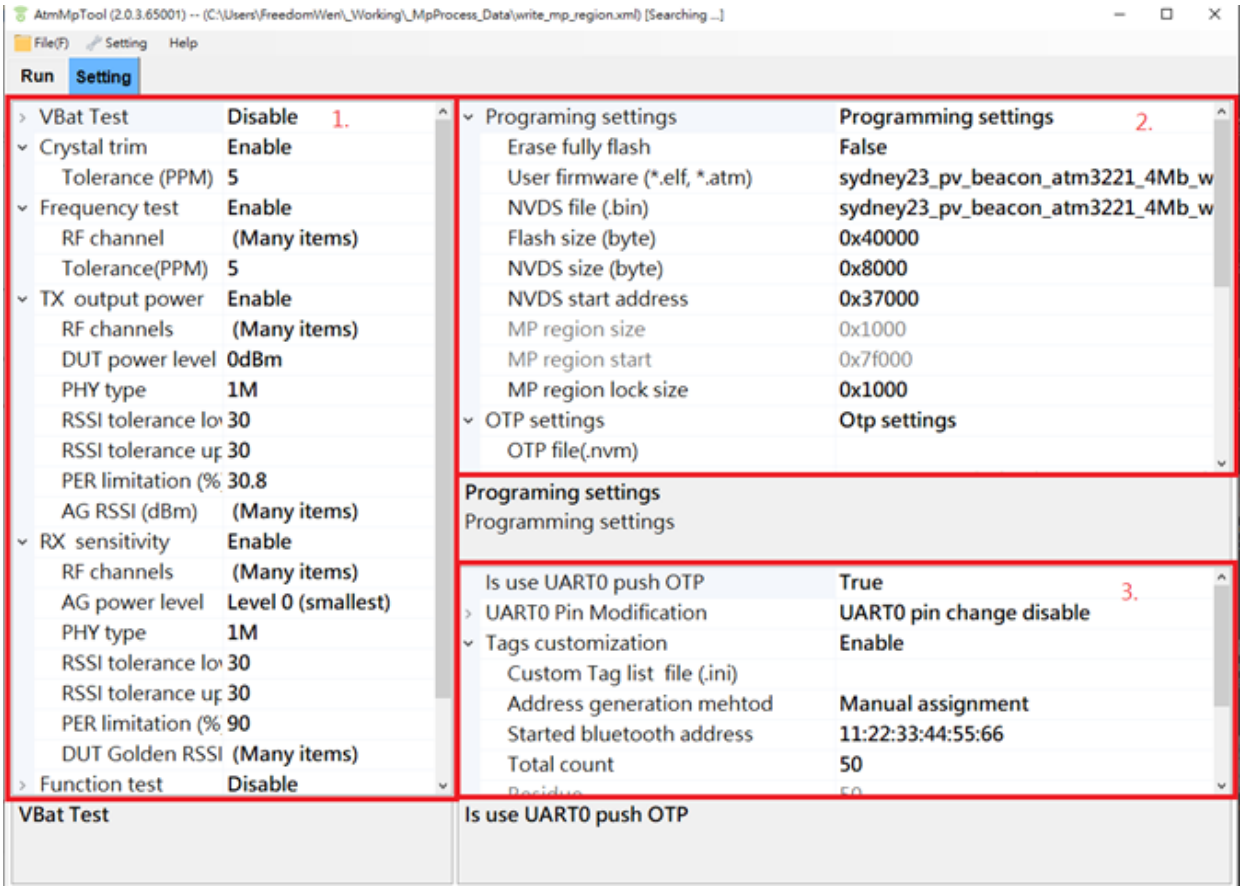


Figure 8-10 Setting Tab

For testing, there are 8 topics:

- Crystal trim
- Frequency test
- TX output power

- RX sensitivity
- Function test
- User firmware(.elf)
- NVDS file (.bin)
- Tags customization
- OTP file (.nvm)

8.4 Crystal Trim

This test item trims the crystal offset by measuring the frequency offset of the crystal signal using APT. After trimming, users can write the crystal CAP (capacitance) to OTP. When performing 'Crystal Trim', the APT will tune the capacitance of the crystal circuit and measure the frequency offset by APT. Figure 8-11 shows the Crystal Trim Setting.

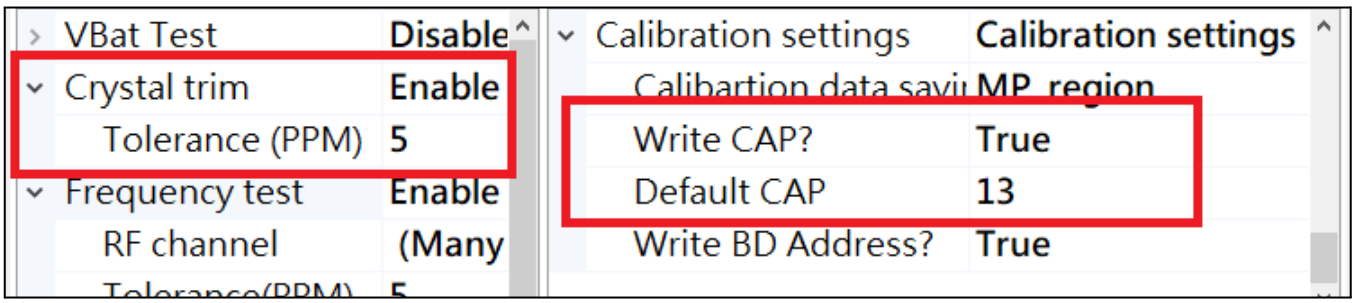


Figure 8-11 Crystal Trim Setting

The 'Tolerance(PPM)' is the accuracy requirement of crystal trim.

When 'CAP to OTP?' set to true, the final CAP value will be written into OTP.

The 'Default CAP' indicates crystal design value. This value will affect the decision of whether to trim or write to OTP. If 'Default CAP' is equal to current CAP value inside the chip, the crystal trim will be skipped. If the trimmed CAP is equal to 'Default CAP', the final CAP OTP written will be skipped.

See Figure 8-12 for test status.

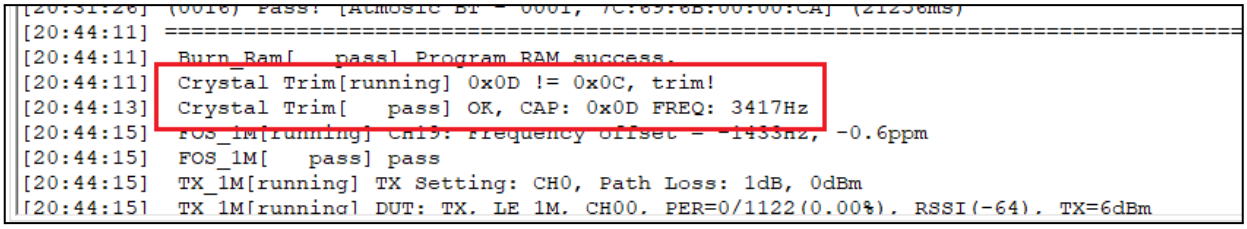


Figure 8-12 Crystal Trim Test Status

8.5 Frequency Test

This test measures the frequency offset of the radio. Figure 8-13 shows the Frequency Test Setting.

The 'Tolerance(PPM)' is the accuracy criteria by percentage per million (PPM) for passing this test.

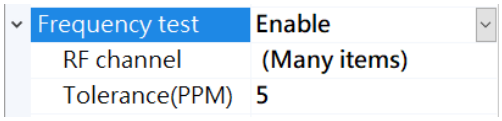


Figure 8-13 Frequency Test Setting

After starting the test, DUT will start to transmit continuous TX, and APT will measure the frequency of DUT. The DUT's frequency must be in the target frequency ± 'Tolerance(PPM)'.

Figure 8-14 shows the test status.

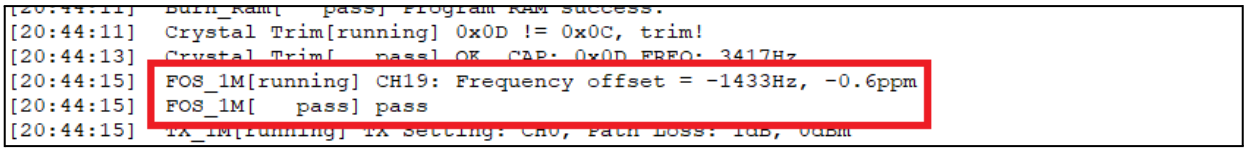


Figure 8-14 Frequency Test Status

8.6 TX Output Power

DUT transmits continuous TX packets on the specified ‘RF channels’, ‘DUT power’ and ‘PHY type’, and the APT measures the DUT RSSI and calibrates DUT’s output power. Figure 8-15 shows the TX Output Power Setting.

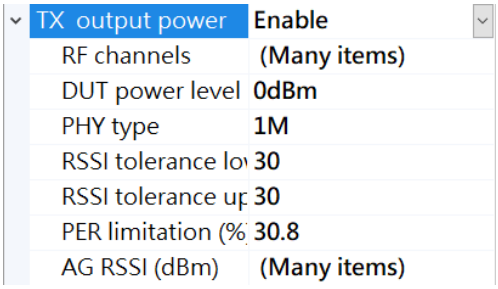


Figure 8-15 TX Output Power Setting

The test result should be in the range of ‘DUT power’ ± ‘Tolerance(db)’. Otherwise, it will report failure. Figure 8-16 shows the TX Output Power Status.

```
[21:05:38]  TX_1M[running] TX Setting: CH0, Path Loss: 1dB, 0dBm
[21:05:39]  TX_1M[running] DUT: TX, LE_1M, CH00, PER=3/1154(0.26%), RSSI(-65), TX=5dBm
[21:05:39]  TX_1M[running] TX Setting: CH19, Path Loss: 1dB, 0dBm
[21:05:40]  TX_1M[running] DUT: TX, LE_1M, CH19, PER=0/1176(0.00%), RSSI(-66), TX=3dBm
[21:05:40]  TX_1M[running] TX Setting: CH39, Path Loss: 1dB, 0dBm
[21:05:41]  TX_1M[running] DUT: TX, LE_1M, CH39, PER=0/1114(0.00%), RSSI(-67), TX=1dBm
[21:05:41]  TX_1M[   pass] Success
```

Figure 8-16 TX Output Power Status

8.7 RX Sensitivity

RX Sensitivity tests the PER of DUT. APT transmits burst TX on specified 'RF channels', Atmosic Golden IC 'AG power level' and 'PHY type', and DUT starts to receive for a certain time. Figure 8-17 shows the RX Sensitivity Setting.

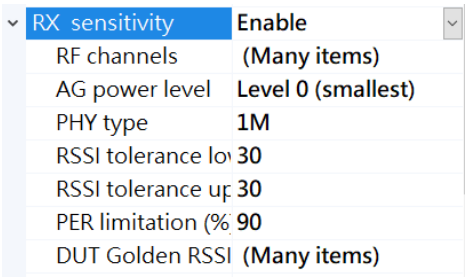


Figure 8-17 RX Sensitivity Setting

If the PER is lower than 30.8%, RX sensitivity passes. Otherwise it will report failure. Figure 8-18 shows the RX Sensitivity Status.

```
[21:05:41]  RX_1M[running] RX Setting: CH0 Power:Level 1, Path Loss: 1dB
[21:05:42]  RX_1M[running] DUT:RX, LE_1M, CH00, PER=0/1198(0.00%), DUT RSSI(-72).
[21:05:42]  RX_1M[running] RX Setting: CH19 Power:Level 1, Path Loss: 1dB
[21:05:43]  RX_1M[running] DUT:RX, LE_1M, CH19, PER=0/1069(0.00%), DUT RSSI(-73).
[21:05:43]  RX_1M[running] RX Setting: CH39 Power:Level 1, Path Loss: 1dB
[21:05:44]  RX_1M[running] DUT:RX, LE_1M, CH39, PER=7/1213(0.58%), DUT RSSI(-75).
[21:05:44]  RX_1M[   pass] Success
```

Figure 8-18 RX Sensitivity Status

8.8 Function Test

Function test is a test for normal Bluetooth LE operations. There are two types of testing: advertisement and connection by selecting ‘Advertising’ or ‘Connection’ to true respectively. Figure 8-19 shows the Function Test Setting.

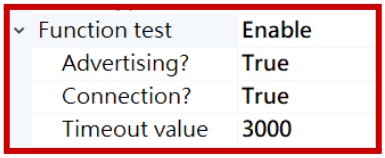


Figure 8-19 Function Test Setting

The ‘Timeout value’ is the time within which APT could complete the operation. Otherwise it will report failure. This test will launch after the normal firmware is programmed, if the firmware doesn’t support advertising this item should not be enabled.

Figure 8-20 shows the Function Test Status.

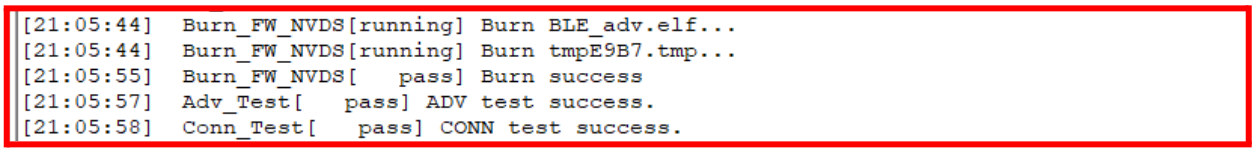


Figure 8-20 Function Test Status

8.9 User Firmware (.atm, .elf) / NVDS file (.bin) / Tags Customization

After selecting files to the ‘User firmware (.elf)’ and ‘NVDS file (.bin)’, AtmMPtool will program flash after all RF related test items have passed. Figure 8-21 shows the User Firmware/Tags Setting.

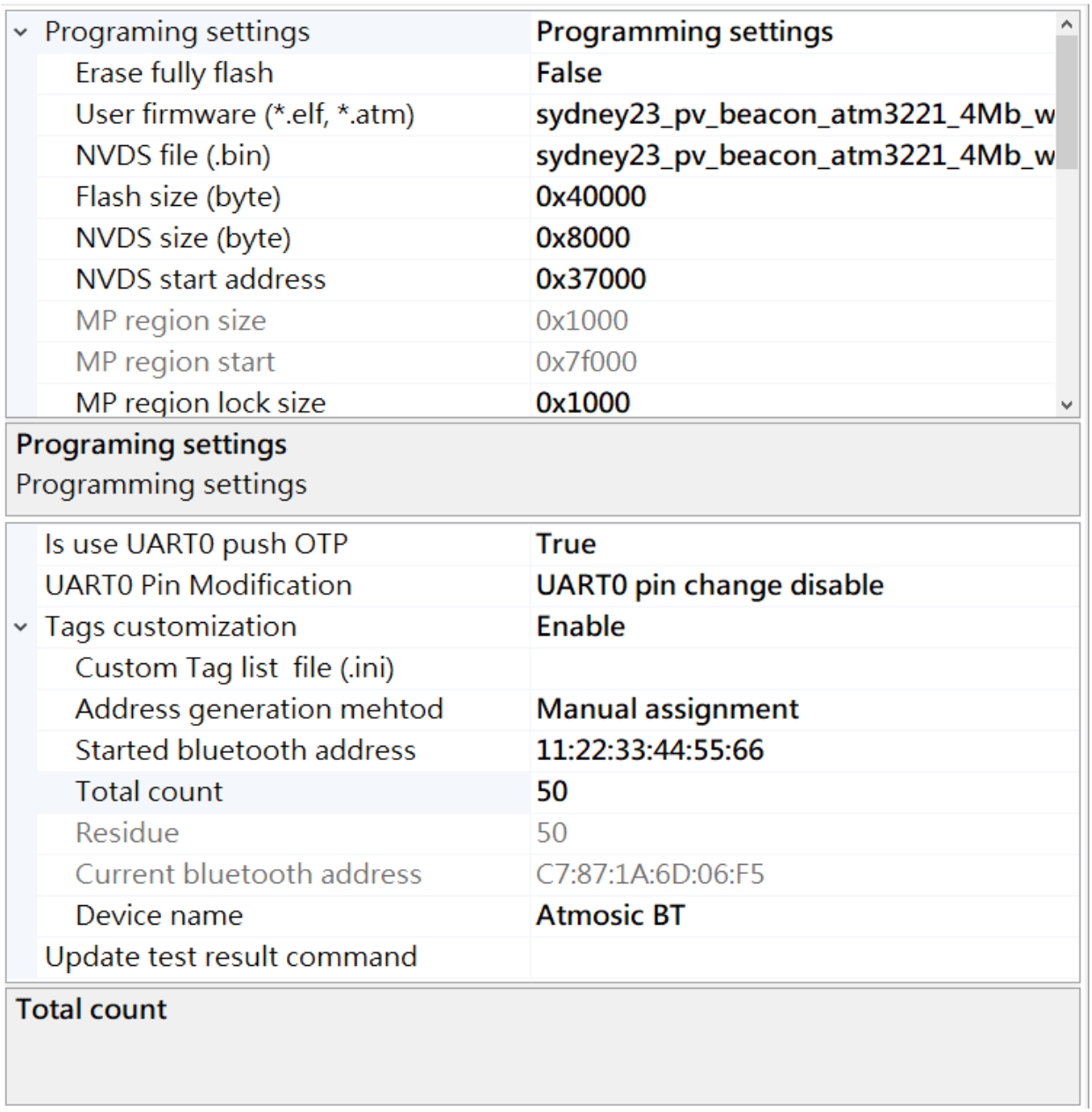


Figure 8-21 User Firmware/Tags Setting

The Tags customization is used to give every device a different BT address, name (if needed) and application specific tag. For the application specific tag, it could be defined by assigning ‘Custom Tag List file’. The format of ‘Custom Tag List file’ is described in Figure 8-22.

```
<Number of Entry(decimal)> = <Tag(hex)>
<Length(hex)>
0=<Data(hex)>
1=<Data(hex)>
2=...
...
```

Figure 8-22 Format of ‘Custom Tag List file’

Figure 8-23 shows 500 entries with tag 0xAA and its data length is 48 bytes.

```
500 = AA 30
0=e5a72e4629d341389dbe26eef485d08ab643f236356342bfa3b4f63c7cbc6558bf4f7c6a56394a018cfecac61078755f
1=646c921c03ac48bfbab22230fd36760f7e3c32f08226434ea50c3ec8ff6f3b6c32109069ce0042329e70bf4ee8b5a7c8
2=f7791bac37dc4c4ebd8882c160fd17b294e7cf1c52e94642810014ccf33d883ec1873d1ff6214239bfa3ebadc9c1a6f0
3=...
```

Figure 8-23 Example of 500 Entries

For BT address tag, there are two kinds of address generation methods: manual assignment and auto generate address. If auto generate address mode is selected, the address will be generated randomly as a static random address of Bluetooth standard on each device. If manual assignment is selected, the user needs to define address range as an address pool. The BT address pool which is started from ‘Started Bluetooth address’ and has ‘Total count’ number of addresses. These addresses will be assigned to each tested DUT. The ‘Residue’ indicates the number of unused addresses in the pool. The ‘Current Bluetooth address’ indicates the next address which will be assigned to the next DUT.

If the pool is empty when the addresses run out, some messages will be shown to the user and AtmMPtool will not assign an address to DUT until the user specifies a new address pool. Figure 8-24 shows Out of Address Error.

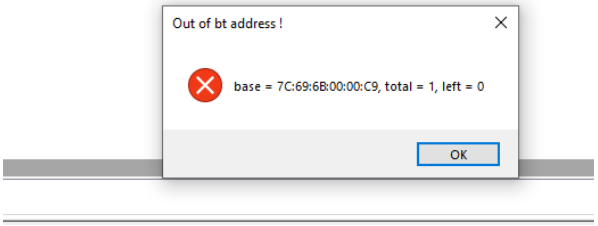


Figure 8-24 Out of Address Error

The ‘Device name’ is used to specify the DUT’s name. It could be treat with a c# string with an extra parameter of uint64 form of assigned BT address. For example, if the current DUT BT address is 7C:69:6B:00:03:FF and ‘Device name’ is AT-{0:X3}, then the result would be AT-3FF.

The ‘BDADDR to OTP’ property gives users a way to store BT addresses into OTP.

8.10 OTP File (.nvm)

This property is used to assign a specific NVM file to update OTP. After selecting the file, the OTP area will be programmed in the last step of the MP process. Usually, the NVM file is used for OTP application or some complement HW parameters setting.

8.11 UART0 Pin Modification

In the Setting page, under the option “UART0 pin Modification”, please set the UART pins according to the GPIO used on the board. See Figure 8-25.

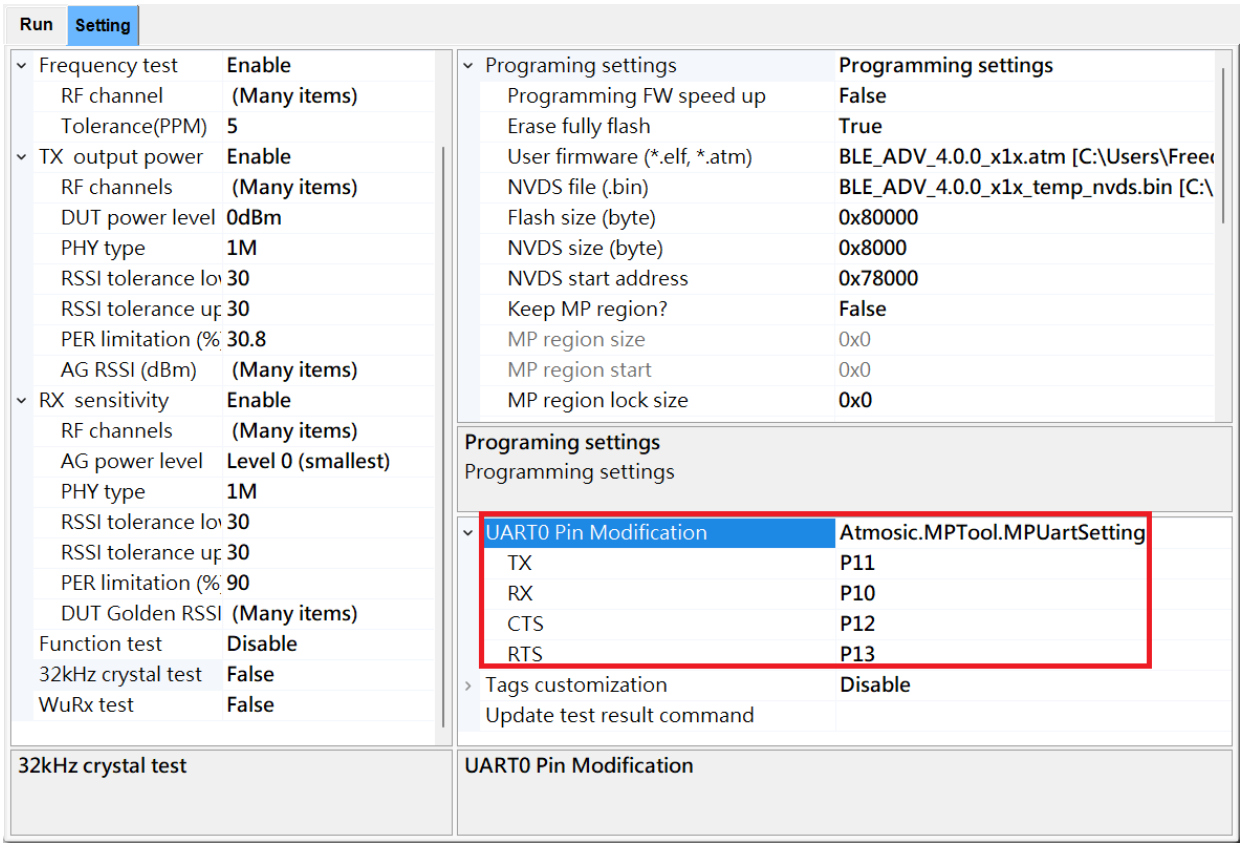


Figure 8-25 UART0 Pin Modification

If your board is Atmosic's EVK Table 8-1 shows the UART setting for EVBs and modules:

Table 8-1 UART Setting for EVB and Module

	EVB-5x5 mm Package	EVB-6x6 mm Package	ATM22-MOD Module	ATM32-MOD Module
TX	P30	P11	P23	P23
RX	P25	P10	P25	P25
RTS	P13	P13	P24	P24
CTS	P11	P12	P11	P11

8.12 Start Option

The ‘Start Option’ property is for users to choose how to start the MP process. Currently, two options are supported, ‘Button Start’ and ‘SN Input Start’. When ‘Button Start’ is selected, the user clicks the ‘Start/Stop’ button to control the testing procedure. When ‘SN Input Start’ is selected, the user inputs the serial number into the SN text box then press enter to start. The serial number default is 12 heximal letters. Figure 8-26 shows the selection of start options. Figure 8-27 shows the SN text box which waits for start.

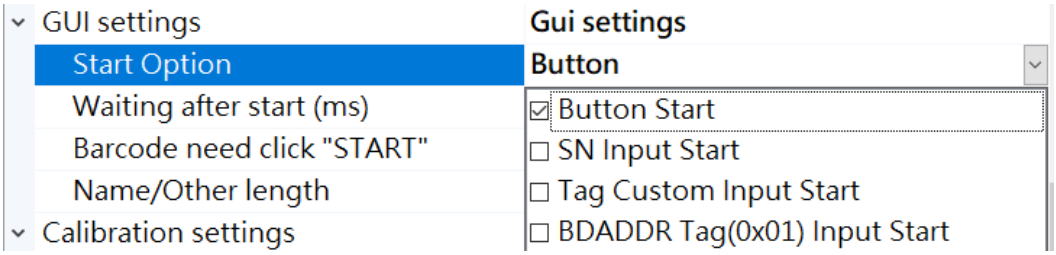


Figure 8-26 Start Option

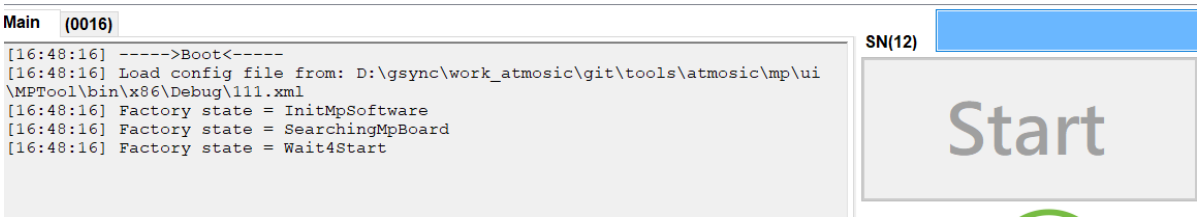


Figure 8-27 SN Text Box

8.13 Write MP Region

If your firmware supports calibration data to be written in the MP region, you can write calibration data in it through the following settings.

When you load an atm file of the support MP region, you will see the fields in Figure 8-28 display the MP region information corresponding to the atm file.

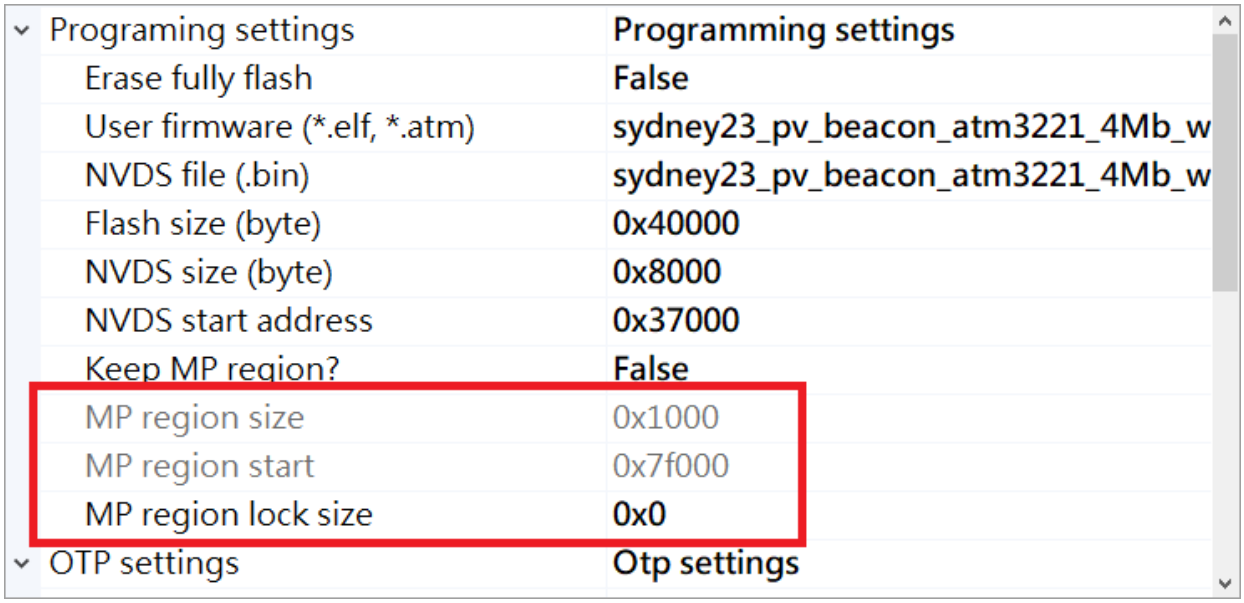


Figure 8-28 MP Region Information

You also need to select this option to set up the MP process to write calibration data into the MP region.

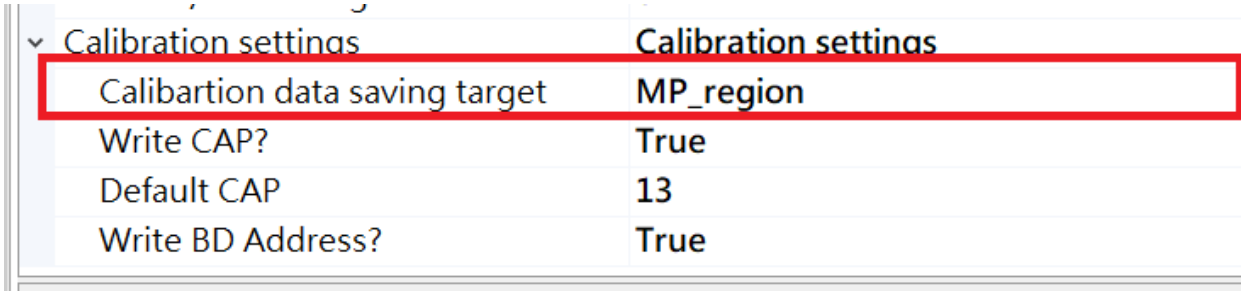


Figure 8-29 Write Calibration Data to MP Region

And choose which data need to write:

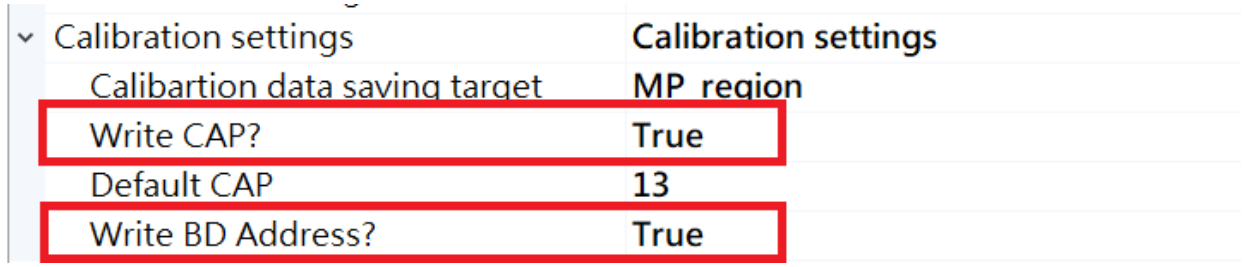


Figure 8-30 Choose Which Calibration Data to Write

8.14 Command Mode

After clicking the menu item ‘Command mode’, a console will show up. Figure 8-31 shows the Command Mode Setting.

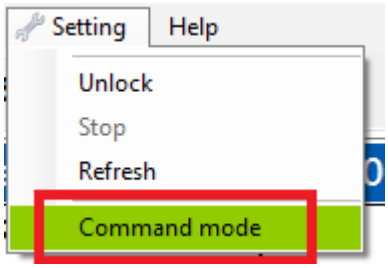


Figure 8-31 Command Mode Setting

By typing any invalid word, the usage will display to guide user usage. Figure 8-32 shows the Command Menu.

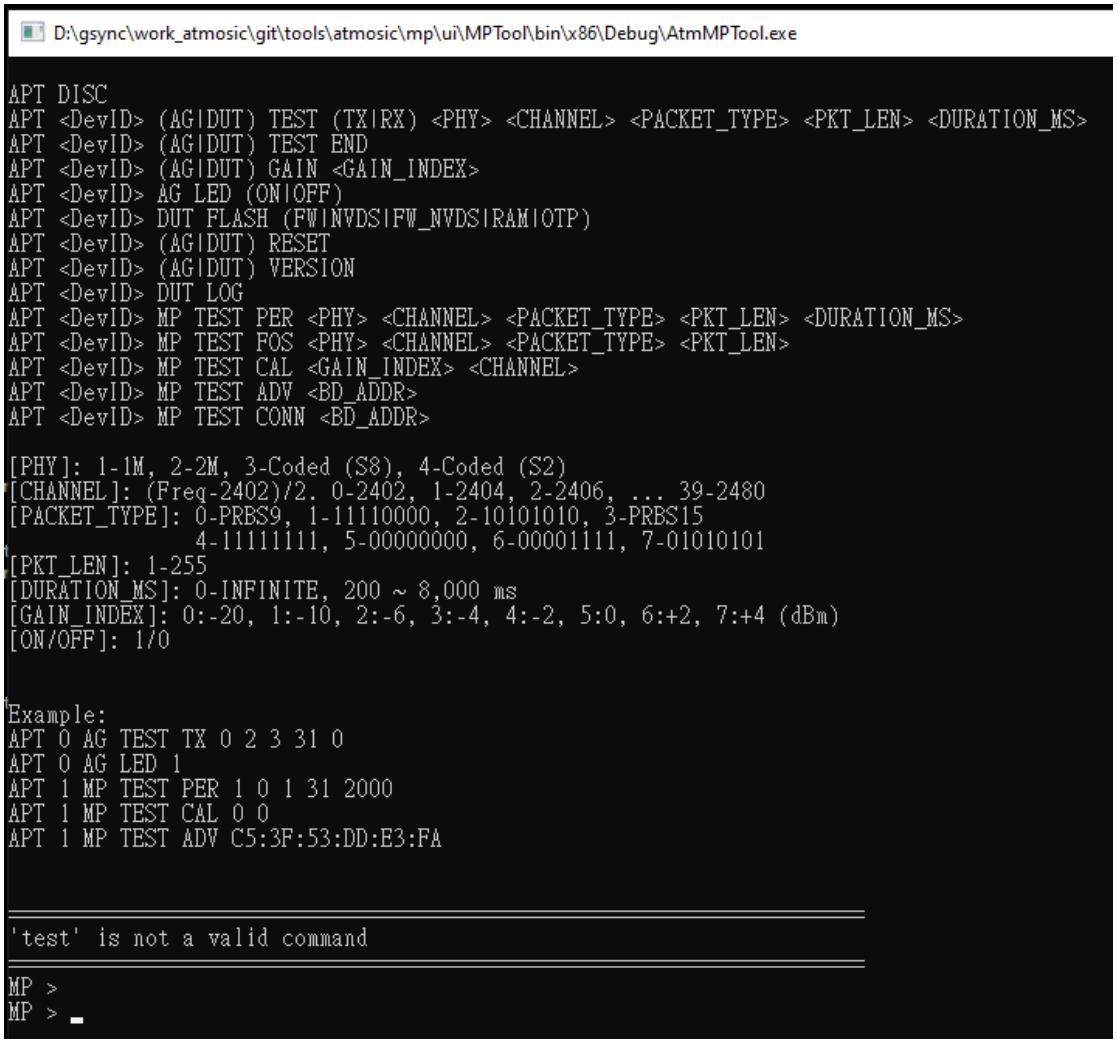


Figure 8-32 Command Menu

Users can use command mode to test against a single APT.

9. Atmosic Production Tool Console Mode

The Atmosic Mass Production Tool also provides console mode for user control. The user needs to go to the working directory in “C:\AtmosicMP\bin” and input the command. Figure 9-1 shows all commands that are supported by the Atmosic Mass Production Tool console mode and the user should follow the command format “AtmMPtool APT parameters...”.

There are few conventions in these command formats:

- “S” symbol is a fixed string S. The matched input could be case-insensitive.
- “<S>” symbol is for dynamic inputs, like id and file name. The matched input would depend on the situation of the real execution environment.
- “(S1|S2|...|Sn)” symbol is one of the fixed strings between S1 to Sn. The matched input shall be one of S1 to Sn with case-insensitive.
- “[S]” symbol is for enumerated number. Only the predefined number allowed.
- “<<S>>” symbol is optional and the definition is the same as “<S>”.
- “[[S]]” symbol is optional and the definition is the same as “[[S]]”.

1	2	3	4	5	6	7	8	9	10	11	12
APT	DISC										
APT	<DevID>	(AG|DUT)	RESET								
APT	<DevID>	(AG|DUT)	VERSION								
APT	<DevID>	(AG|DUT)	NVDS	READ	TAG						
APT	<DevID>	(AG|DUT)	TEST	(TX|RX)	[PHY]	[CHNL]	[PKTYPE]	<PktLen>	<Tms>		
APT	<DevID>	(AG|DUT)	TEST	END							
APT	<DevID>	(AG|DUT)	GAIN	[GAINIDX]							
APT	<DevID>	AG	LED	(ON|OFF)							
APT	<DevID>	DUT	FLASH	RAM	<<file>>						
APT	<DevID>	DUT	FLASH	(OTP|POTP)	<file>						
APT	<DevID>	DUT	FLASH	N32C							
APT	<DevID>	DUT	FLASH	NVDS|PNVDS	<<bin>>	<<addr>>	<<size>>				
APT	<DevID>	DUT	FLASH	FW	<elf>	<<addr>>	<<size>>	[[EUD]]			
APT	<DevID>	DUT	FLASH	VFW	<elf>	<<addr>>					
APT	<DevID>	DUT	FLASH	BOTH	<elf>	<bin>	<<faddr>>	<<fsize>>	<<naddr>>	<<nsize>>	[[EUD]]
APT	<DevID>	DUT	FLASH	ERASE	<faddr>	<fsize>					
APT	<DevID>	DUT	UART	MUX	<Tx>	<Rx>	<Cts>	<Rtx>			
APT	<DevID>	DUT	LOG								
APT	<DevID>	MP	TEST	PER	[PHY]	[CHNL]	[PKTYPE]				
APT	<DevID>	MP	TEST	FOS	[PHY]	[CHNL]	[PKTYPE]				
APT	<DevID>	MP	TEST	CAL	[GAINIDX]	[CHNL]					
APT	<DevID>	MP	TEST	ADV	<bdaddr>			<PktLen>	<TmsTx>	<TmsRx>	
APT	<DevID>	MP	TEST	CONN	<bdaddr>			<PktLen>			
FILE	(OTP|NVDS)	<file>	SHOW								
FILE	(OTP|NVDS)	<file>	TAG	ADD	<tag>	<hex>	<new file>				
FILE	(OTP|NVDS)	<file>	MERGE	<add file>	<new file>						
HELP	(MP|DUT|AG|FILE)										

Figure 9-1 Command List

Before OTP and NVDS programming, users may need to prepare an updated file. The Atmosic Mass Production Tool provides a tool named nvds_tool for users to manipulate the tags and form the burned file. Please refer to the OTP/NVDS/MP Region File Content Display section for the usage.

9.1 System Environment Variable

This chapter shows some system environment variables used to adjust the CLI test:

Variable name	Description
ROM_VERSION	Please set this environment by your IC: • ATMx2xx-x0x → set ROM_VERSION=x0x • ATMx2xx-x1x → set ROM_VERSION=x1x • ATM33xx-3 → set ROM_VERSION=33-3 • ATM33xx-5 → set ROM_VERSION=33-5
SKIP_DETAIL_LOG	If you want to simplify the log generated by the CLI in the Console, you can set this environment variable to 1.

9.2 Board Discovery

Discover the connected APT/DL boards by using the following command, also see Figure 9-2.

Command	
AtmMPTool APT DISC	
Parameter:	
DISC	Discover APT boards.
Result	Return all <DevID> of APT boards

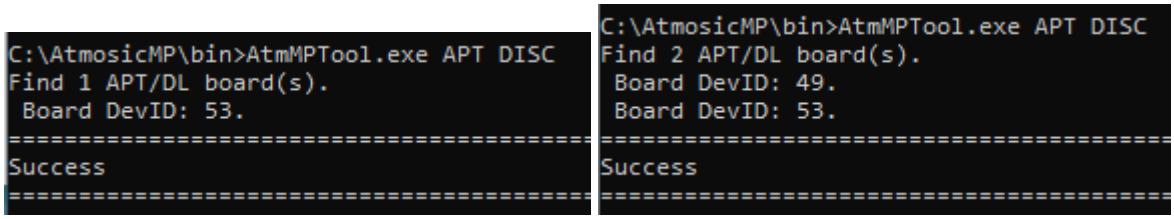


Figure 9-2 Board Discovery

9.3 DUT/AG Tx Testing

Apply the AG or DUT to Bluetooth LE Tx mode. Also see Figure 9-3.

Command	
AtmMPTool APT <DevID> (AG|DUT) TEST [TX] [PHY] [CHNL] [PKTYPE] <PktLen> <Tms>	
Parameter:	
(AG|DUT)	AG - Atmosic golden. (Not allowed with DL board) DUT - Device under test.
[TX]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[PKTYPE]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15

	4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 255
<Tms>	How long will keep transmitting packets 1 ~ 8000 0: Infinite
Result	Return how many packets transmitted.

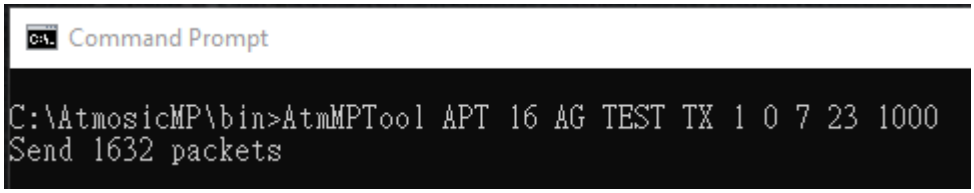


Figure 9-3 AG TX Test

***Note**: Users must download HCI(DTM) firmware to RAM prior to using this command with DUT (AG|DUT).*

9.4 DUT/AG Rx Test

Apply the AG or DUT to Bluetooth LE Rx mode. See also Figure 9-4.

Command
AtmMPTool APT <DevID> (AG|DUT) TEST RX [PHY] [CHNL] [PKTYPE] <PktLen> <Tms>
AtmMPTool APT <DevID> (AG|DUT) TEST END

Parameter:	
(AG|DUT)	AG - Atmosic golden. (Not allowed with DL board) DUT - Device under test.
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15

	4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 255
<Tms>	Any number from 1 to 8000 but cannot set 0.
Result	Return how many packets received.

***Note**: Users must download HCI(DTM) firmware to RAM prior to using this command with DUT (AG|DUT).*

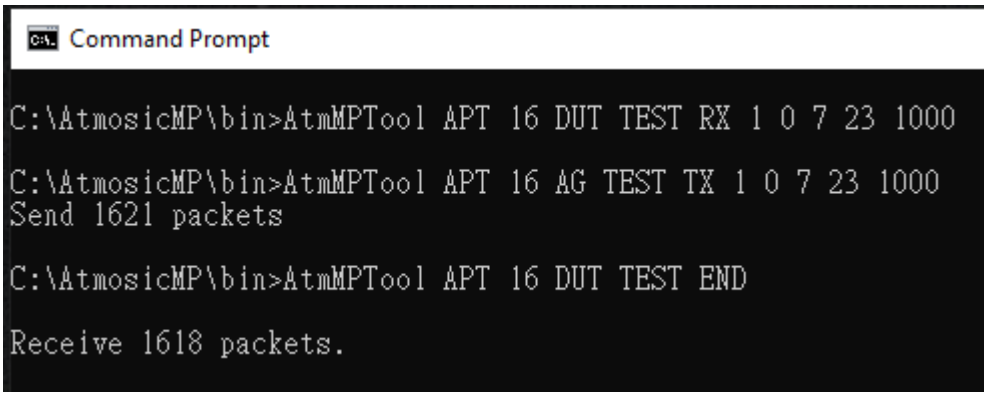


Figure 9-4 DUT RX Test

9.5 DUT/AG Tx Power Gain

Command and parameters for setting the Tx power to AG or DUT.

Command	
AtmMPTool APT <DevID> (AG|DUT) GAIN [GAINIDX]	
Parameter:	
(AG|DUT)	AG - Atmosic golden. (Not allowed with DL board) DUT - Device under test.
[GAINIDX]	0 - (-20dBm) 1 - (-10dBm) 2 - (-6dBm) 3 - (-4dBm) 4 - (-2dBm) 5 - 0dBm

	6 - 2dBm 7 - 4dBm

Note: Users must download HCI (DTM) firmware to RAM prior to using this command with DUT role.

9.6 DUT RAM or OTP Programming

Program DUT RAM FW or DUT OTP data. Also see Figure 9-5.

Notice: If you downloaded default RAM code and want to use UART, please call UART rx enable command or change RX pin mux first.

Command
AtmMPTool APT <DevID> DUT FLASH (RAM|OTP|POTP|NSWD|N32C) <file>

Parameter:	
(RAM|OTP|POTP|N32C)	RAM - Program image to RAM for RF testing OTP - Program OTP data to OTP **(CANNOT REVERT SETTING!)** POTP - Pull OTP data from OTP NSWD - SWD connection will be prohibited after setting **(CANNOT REVERT SETTING!)** N32C - No 32 KHz crystal on board **(CANNOT REVERT SETTING!)**
<file> Optional for RAM Not used in N32C	For RAM: If not provided, C:\AtmosicMP\fw\DTM.bin will be used. For OTP : Read this file and program to OTP. For POTP: Pull data from OTP and write to this file.

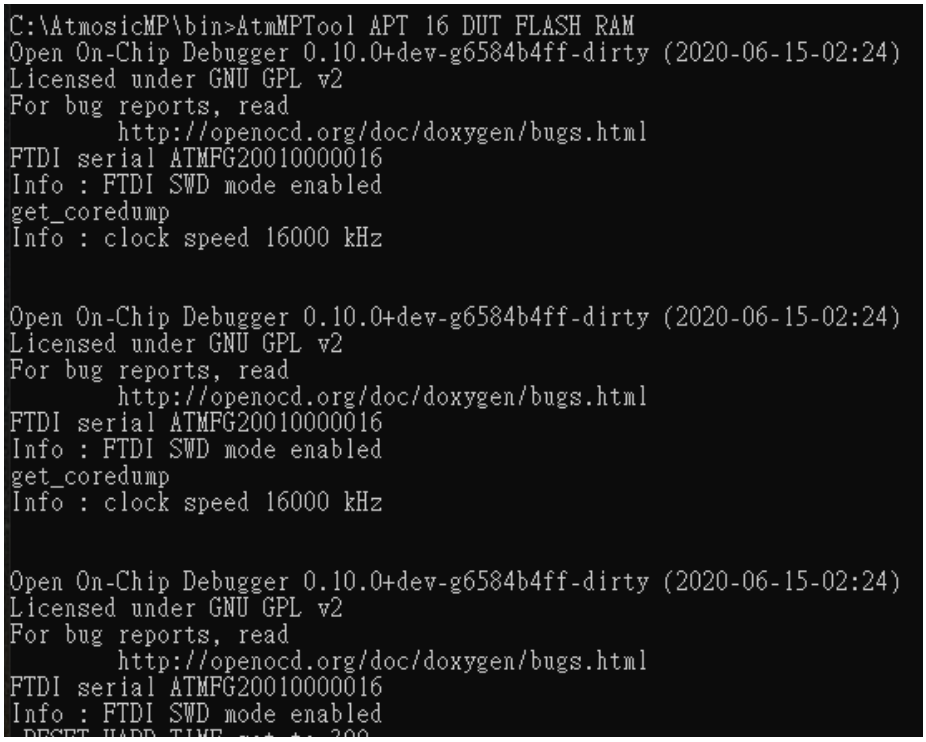


Figure 9-5 RAM Programming

9.7 DUT Firmware Programming

Program firmware to flash. Also see Figure 9-6.

Command

- AtmMPTool APT <DevID> DUT FLASH FW <atm file> [EUD] [NCE]
- AtmMPTool APT <DevID> DUT FLASH FW <elf file> <addr> <size> [EUD] [NCE]

Parameter	
FW	FW - Program firmware and verification.
<atm file>, <elf file>	File pathname. (.elf), E.g, C:\\AtmosicMP\\Test.elf for firmware.
<addr> Optional	FW start address. This information is customer specific. If not provided, the default address of FW is 0x0 respectively.
<size> Optional	FW maximal size. This information is customer specific. If not provided, the default maximum size of FW is 0x80000. (512 KB).
[EUD] Optional	Erase upgrade data or not. 1 for erasing. If not provided, the default value is 1.
[NCE]	Do chip erase before programing flash

Optional	
Result	No error message

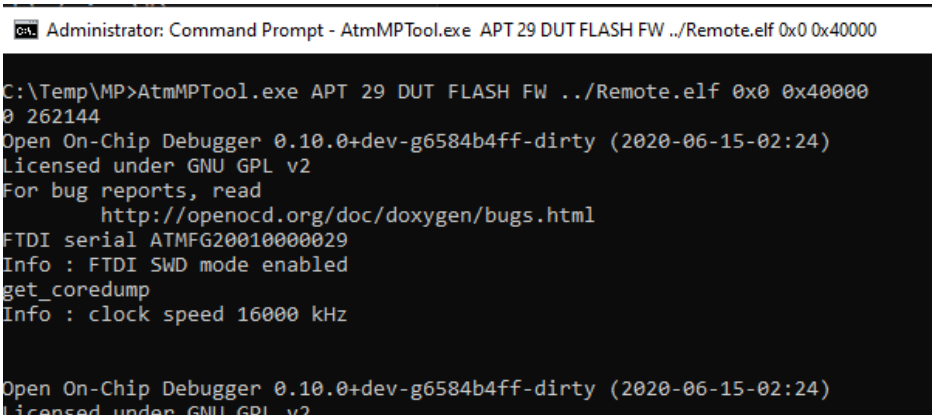


Figure 9-6 Firmware Programming

9.8 DUT NVDS Programming or Pull

Program NVDS data to flash. Also see Figure 9-6.

Command	
• AtmMPTool APT <DevID> DUT FLASH (NVDS|PNVDS) <file> <addr> <size> • AtmMPTool APT <DevID> DUT FLASH (NVDS|PNVDS) <file>	
Parameter	
(NVDS|PNVDS)	NVDS - Program NVDS data to flash. PNVDS - Pull nvds data from FLASH
<file>	NVDS file pathname. (.bin). E.g, C:\\AtmosicMP\\flash_nvds.bin.
<addr> Optional	NVDS start address. This information is customer specific. If not provided, the default address of NVDS is 0x78000.
<size> Optional	NVDS maximal size. This information is customer specific. If not provided, the default maximum size of NVDS is 0x8000. (32 KB).
Result	No error message

9.9 DUT Firmware Validation

Validate firmware in flash.

Command
• AtmMPTool APT <DevID> DUT FLASH VFW <file> <addr> • AtmMPTool APT <DevID> DUT FLASH VFW <file>

Parameter	
<file>	File to be programmed. ELF file for firmware. For Example, C:\\AtmosicMP\\Test.elf
<addr> optional	FW start address. This information is customer specific. If it is not defined, the default value is 0x0.
Result	No error message

9.10 DUT Firmware and NVDS Programming Concurrently

Program firmware and NVDS data to flash at the same time.

Command
• AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN> <FADR> <FSIZE> <NADR> <NSIZE> <EUD> • AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN> <FADR> <FSIZE> <NADR> <NSIZE> • AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN> <FADR> <FSIZE> • AtmMPTool APT <DevID> DUT FLASH BOTH <ELF> <BIN>

Parameter	
<ELF>	FW file (.elf) to be programmed. For Example, C:\\AtmosicMP\\Test.elf
<BIN>	NVDS file (.bin) to be programmed. For Example, C:\\AtmosicMP\\flash_nvds.bin
<FADR> Optional	FW start address. This information is customer specific. If it is not defined, the default value is 0x0.
<FSIZE> Mandatory, if set FADR	FW max size.This information is customer specific. If it is not defined, the default value is 0x80000 (512 KB).
<NADR> Optional	NVDS start address. This information is customer specific. If it is not defined, the default value is 0x78000.

<NSIZE> Mandatory, if set ADDR	NVDS max size. This information is customer specific. If it is not defined, the default value is 0x8000 (32 KB).
<EUD> Optional	Erase upgrade data or not. 1 for erasing. If not provided, the default value is 1.
Result	No error message

9.11 DUT Flash Erase

Program firmware and NVDS data to flash at same time.

Command
• AtmMPTool APT <DevID> DUT FLASH ERASE <FADR> <FSIZE>

Parameter	
<FADR>	Start address. If not provided, default is 0.
<FSIZE>	Erase Size. If not provided, the default value will be the flash size.

9.12 DUT/AG Reset Command

Reset AG or DUT.

Command
AtmMPTool APT <DevID> (AG|DUT) RESET

Parameter	
(AG|DUT)	AG - Atmosic golden. (Not allowed with DL board) DUT - Device under test.
Result	No error message

9.13 DUT/AG Version

Command to query the chip version of AG or DUT. Also see Figure 9-7.

Command	
AtmMPTool APT <DevID> (AG|DUT) VERSION	
Parameter	
(AG|DUT)	AG - Atmosic golden. (Not allowed with DL board) DUT - Device under test.
Result	Show chip version of AG or DUT.

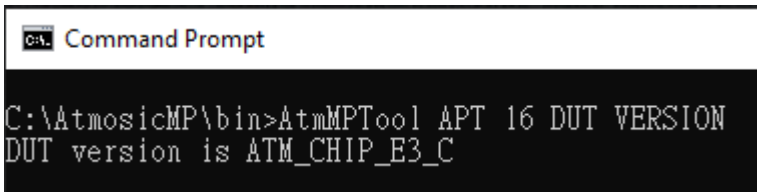


Figure 9-7 Chip Version

***Note**: Users must download HCI (DTM) firmware to RAM first prior to using this command with the DUT role.*

9.14 DUT/AG NVDS Read

Read NVDS tag data. Also see Figure 9-8.

Command	
AtmMPTool APT <DevID> (AG|DUT) NVDS READ <TAG>	
Parameter	
(AG|DUT)	AG - Atmosic golden. (Not allowed with DL board) DUT - Device under test.
<TAG>	Decimal value from 0 to 255. Please get this value from RD.
Result	Show the NVDS data.

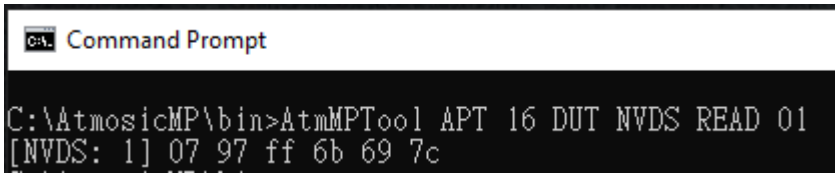


Figure 9-8 NVDS Read

9.15 DUT Log

Command to show the DUT log after DUT power on. See also Figure 9-9.

Command	
AtmMPTool APT <DevID> DUT LOG	
Result	Show DUT log

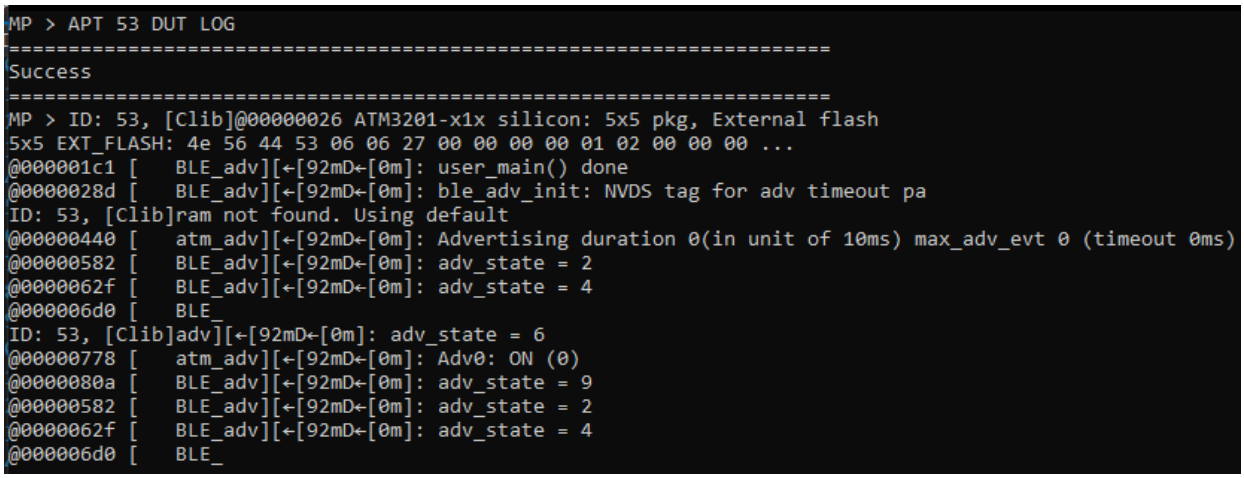


Figure 9-9 DUT Debug Log

9.16 Bluetooth LE RF PER Test (Packet Error Rate)

Command for packet error rate testing. See also Figure 9-10.

Command	
AtmMPTool APT <DevID> MP TEST PER [PHY] [CHNL] [PKTYPE] <PktLen> <Tms>	
Parameter	
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15 4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 63
<Tms>	Set the PER test duration. 200 ~ 8000
Result	Show the PER test result with both directions.

***Note**: Users must download HCI(DTM) firmware to RAM prior to using this command.This command is only allowed with the APT board, not with DL board.*

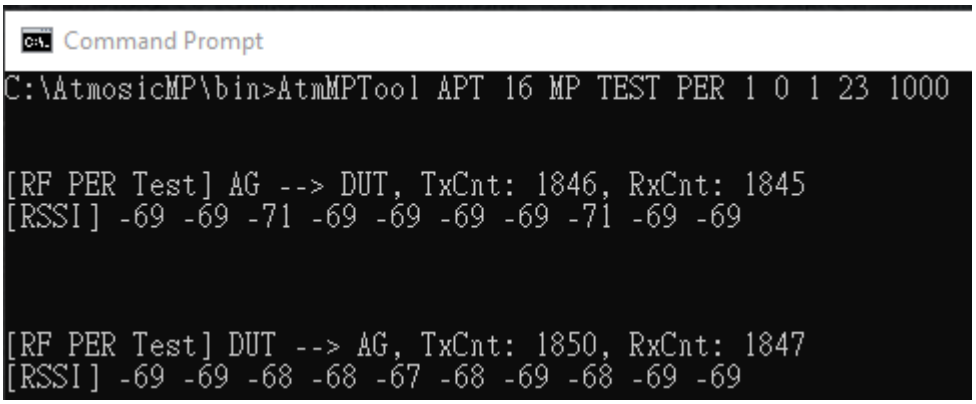


Figure 9-10 PER Test With AG (APT Only)

9.17 Bluetooth LE RF FOS Test (Frequency Offset)

Command for frequency offset testing. See also Figure 9-11.

Command	
AtmMPTool APT <DevID> MP TEST FOS [PHY] [CHNL] [PKTYPE] <PktLen>	
Parameter	
[PHY]	1 - 1M 2 - 2M 3 - Coded (S8) 4 - Coded (S2)
[CHNL]	0 ~ 39
[PKTYPE]	0 - PRBS9 1 - 11110000 2 - 10101010 3 - PRBS15 4 - 11111111 5 - 00000000 6 - 00001111 7 - 01010101
<PktLen>	0 ~ 63
Result	Show the FOS test result with both directions.

***Note**: Users must download HCI (DTM) firmware to RAM prior to using this command. This command is only allowed with the APT board, not with DL board.*

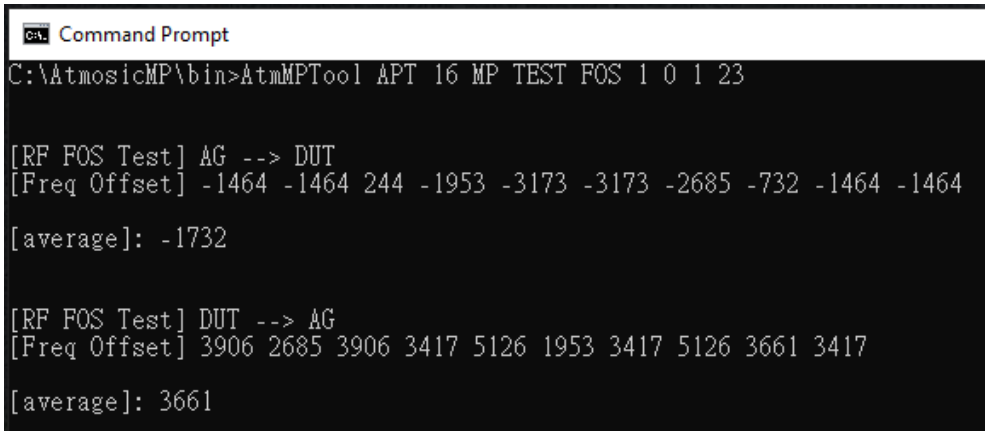


Figure 9-11 FOS Test (APT Only)

9.18 Bluetooth LE RF CAL Test (Calibration)

Use APT to calibrate M crystal. See also Figure 9-12.

Command	
AtmMPTool APT <DevID> MP TEST CAL [GAINIDX] [CHNL]	
Parameter	
[GAINIDX]	Tx power gain: 0 - (-20 dBm) 1 - (-10 dBm) 2 - (-6 dBm) 3 - (-4 dBm) 4 - (-2 dBm) 5 - 0 dBm 6 - 2 dBm 7 - 4 dBm
[CHNL]	BLE Channel: 0 ~ 39
Result	Show the calibration result.

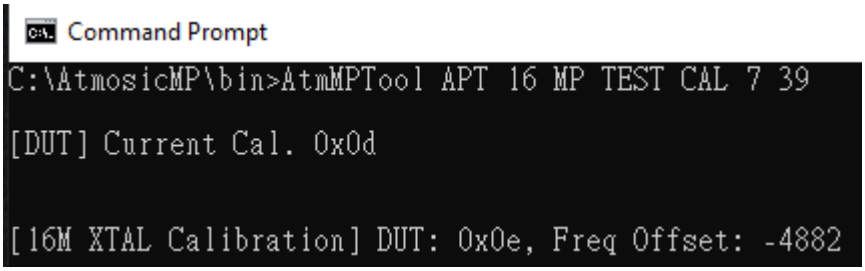


Figure 9-12 RF Calibration (AG Only)

Note: Users must download HCI (DTM) firmware to RAM prior to using this command. This command is only allowed with the APT board, not with DL board.

9.19 OTP/NVDS/MP Region File Content Display

This command shows the content of OTP (.nvm) and NVDS (.bin) files. See also Figure 9-13.

Command	
AtmMPTool FILE (OTP | NVDS | MPREGION) <file> SHOW	
Parameter	
(OTP | NVDS | MPREGION)	OTP - OTP file type NDVS - NVDS file type MPREGION - MP region file type
<file>	File path which used to be read and displayed.
Result	Shows the content of the file with rows of tags.

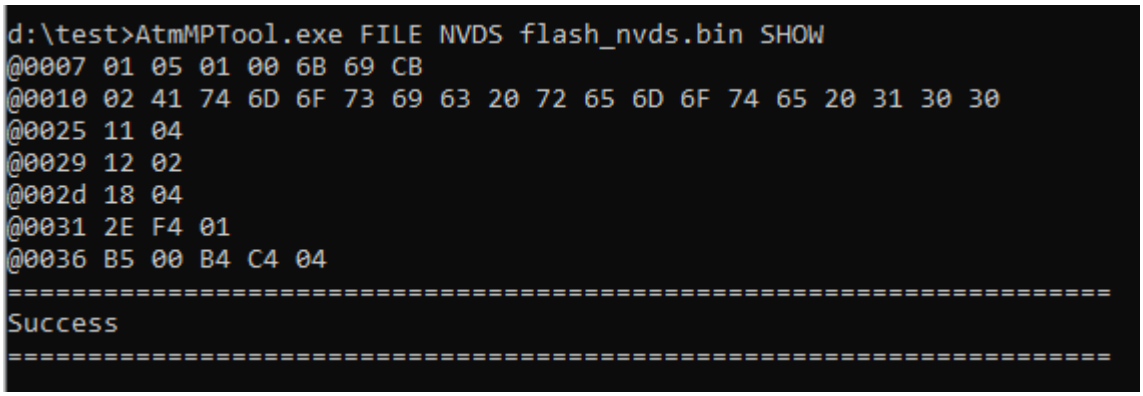


Figure 9-13 OTP and NVDS File Content Display

9.20 NVDS/MP Region Create Blank File

This command can create NVDS (.bin) and MP Region(.bin) files.

Command	
AtmMPTool FILE (NVDS | MPREGION) <file> BLANK	
Parameter	
(NVDS | MPREGION)	NDVS - NVDS file type MPREGION - MP region file type
<file>	File path used to be read and displayed.
Result	Shows the content of the file with rows of tags.

9.21 OTP/NVDS/MP region Tag addition with File

This command adds specific TAG and its data from content of OTP (.nvm) and NVDS (.bin) file and output to a new file. See also Figure 9-14.

For Tag ID please refer to section Common OTP/NVDS/MP Region Tag ID.

Command	
AtmMPTool FILE (OTP | NVDS | MPREGION) <file> TAG ADD <tag> <hex data> <new file>	
Parameter	
(OTP | NVDS | MPREGION)	OTP - OTP file type NDVS - NVDS file type MPREGION - MP region file type
<file>	File path of original file.
<tag>	Tag number. Hex format, e.g, 01, 45, ...
<hex data>	Data with hex format, e.g, 6 bytes data would be input by 112233445566.
<new file>	File path to save.
Result	<new file> will be saved and shows success.

```
d:\test>AtmMPTool.exe FILE NVDS flash_nvds.bin TAG ADD 01 010203040506 nvds_new.bin
========================================================================
Success
========================================================================

d:\test>AtmMPTool.exe FILE NVDS nvds_new.bin SHOW
@0007 01 01 02 03 04 05 06
@0010 02 41 74 6D 6F 73 69 63 20 72 65 6D 6F 74 65 20 31 30 30
@0025 11 04
@0029 12 02
@002d 18 04
@0031 2E F4 01
@0036 B5 00 B4 C4 04
========================================================================
Success
========================================================================
```

Figure 9-14 OTP/NVDS Tag Addition with File

9.22 OTP/NVDS/MP region File merge

This command merges two OTP (.nvm) or NVDS (.bin) files.

Command
AtmMPTool FILE (OTP | NVDS | MPREGION) <file> MERGE <add file> <new file>

Parameter	
(OTP|NVDS|MPREGION)	OTP - OTP file type NDVS - NVDS file type MPREGION - MP region file type
<file>	File path of original file.
<add file>	File path of added file.
<new file>	File path to save.
Result	<new file> will be saved and shows success.

9.23 DUT/AG TEST ADV

Start or stop advertising. See Figure 9-15.

Command

- AtmMPTool APT <DevID> (AG|DUT) TEST ADV START <payloads>
- AtmMPTool APT <DevID> (AG|DUT) TEST ADV STOP

Parameter	
(START|STOP)	Testing start or stop
<payloads>	Use --0x<tag>.(ascii|hex)=data to set payload context. e.g. --0x09.ascii=001122334455
Result	Show the ADV command working status.

```
C:\Users\FreedomWen\_Working\_Code\tools\atmosic\mp\Release>AtmMPTool.exe APT 53 DUT TEST
ADV START --0x09.ascii=001122334455
========================================================================
Success
========================================================================
```

Figure 9-15 Start Advertising

9.24 DUT/AG TEST WuRx

WuRx testing commands, test flow please see Figure 9-16, 9-17 and 9-18.

Command

- AtmMPTool APT <DevID> (AG|DUT) TEST WURX SEND_TX_PACKET
- AtmMPTool APT <DevID> (AG|DUT) TEST WURX RESET_RX
- AtmMPTool APT <DevID> (AG|DUT) TEST WURX CHECK_RX_COUNT <count1_limit> <count2_limit>

Parameter	
(SEND_TX_PACKET | RESET_RX | CHECK_RX_COUNT)	SEND_TX_PACKET: send WURX waveform RESET_RX: reset device to catch WuRx packet CHECK_RX_COUNT: check WuRx packet count
<count1_limit>, <count2_limit>	set limit for count
Result	Show the WuRx command working status and register change log.

```
C:\Users\FreedomWen\_Working\_Code\tools\atmosic\mp\Release>AtmMPTool.exe APT 53 DUT TEST WURX RESET_RX

RUN "set_wurx_params" script

######Call OpenOCD proc: "puts ""\n###set_wurx_param###""; set_wurx_param;"######
Open On-Chip Debugger 0.11.0-g1c55e6642-dirty (2021-07-13-08:38)
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
DEPRECATED! use 'adapter driver' not 'interface'
FTDI serial ATMFG21050000053
Info : FTDI SWD mode enabled
DEPRECATED! use 'adapter speed' not 'adapter_khz'
get_coredump
Info : clock speed 16000 kHz
Info : SWD DPIDR 0x0bb11477
Info : Sydney.cpu: hardware has 4 breakpoints, 2 watchpoints
Info : gdb port disabled

###set_wurx_param###
CMSDK_WRPR1_APB7_CTRL : 0x1
CMSDK_PMU_TOP_PMU11 : 0x9803 -> 0x9803
CMSDK_WRPR1_APB8_CTRL : 0x1
CMSDK_PSEQ_WURX_CONFIG : 0x5
CMSDK_PSEQ_WURX_CONFIG : 0x7
CMSDK_PMU_WURX_WURX0_0 : 0xaaaaaaaa
CMSDK_PMU_WURX_WURX0_1 : 0xaaaaaaaa
CMSDK_PMU_WURX_WURX0_2 : 0xc0f78dc
CMSDK_PMU_WURX_WURX0_3 : 0x20180c04
CMSDK_PMU_WURX_WURX0_4 : 0x3fb27c0f
CALL_OPENOCD_ERROR=0
FINAL_ERROR=0
========================================================================
Success
========================================================================
```

Figure 9-16 Set DUT Reset and Wait for WuRx Packet

```
C:\Users\FreedomWen\_Working\_Code\tools\atmosic\mp\Release>AtmMPTool.exe APT 53 AG TEST WURX SEND_TX_P
ACKET
========================================================================
Success
========================================================================
```

Figure 9-17 Set AG Send WuRx Packet

```
C:\Users\FreedomWen\_Working\_Code\tools\atmosic\mp\Release>AtmMPTool.exe APT 53 DUT TEST WURX CHECK_RX
_COUNT 0 0
RUN "check_wurx_det" script

######Call OpenOCD proc: "puts ""\n###check_wurx_det###""; check_wurx_det 0 0;"######
Open On-Chip Debugger 0.11.0-g1c55e6642-dirty (2021-07-13-08:38)
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
DEPRECATED! use 'adapter driver' not 'interface'
FTDI serial ATMFG21050000053
Info : FTDI SWD mode enabled
DEPRECATED! use 'adapter speed' not 'adapter_khz'
get_coredump
Info : clock speed 16000 kHz
Info : SWD DPIDR 0x0bb11477
Info : Sydney.cpu: hardware has 4 breakpoints, 2 watchpoints
Info : gdb port disabled

###check_wurx_det###
Start check_wurx_det
wurx1_rb : 0x16153cc
cnt_det0 : 0x153
cnt_det1 : 0x58
CALL_OPENOCD_ERROR=0
FINAL_ERROR=0
========================================================================
Success
```

Figure 9-18 Set DUT to Calculate WuRx Packet

9.25 DUT/AG I/O setting

Control GPIO pin action or get GPIO pin status.

Command

- AtmMPTool APT <DevID> (AG|DUT) IO (SETUP|SET) <io> <0|1>
- AtmMPTool APT <DevID> (AG|DUT) IO (GET) <io>

Parameter	
(SETUP | SET | GET)	SETUP: set I/O pin to input(0) or output(1) SET: set output I/O pin to low(0) or high(1) GET: get I/O pin low or high
<io>	number of pin
Result	Show the I/O setting command working status.

9.26 DUT/AG set/get runtime crystal CAP

Set or get runtime crystal CAP value

Command

- AtmMPTool APT <DevID> (AG|DUT) CAP SET <val>
- AtmMPTool APT <DevID> (AG|DUT) CAP GET

Parameter	
(SET | GET)	SET: set output I/O pin to low(0) or high(1) GET: get I/O pin low or high
<val>	0 ~ 31
Result	Show the CAP value setting command working status.

9.27 Set UART Rx enable

When you download default RAM code, and need to use UART, please call this command.

Command

- AtmMPTool APT <DevID> DUT UART RX_ENABLE

Result	Show the UART Rx enable command working status.

9.28 Change UART Pin Mux

If your product's UART pin definition does not equal the default setting, please usel this command to change it before calling UART RX enable.

Command

- AtmMPTool APT <DevID> DUT UART MUX <Tx pin> <Rx pin> <Cts pin> <Rts pin>

Parameter	
<Tx pin><Rx pin> <Cts pin><Rtx pin>	Please key in I/O number
Result	Show the UART pin mux setting command working status.

9.29 Set 16M Crystal CAP

When you need to set CAP value into your OTP or NVDS file, you can use this powerful command to replace OTP/NVDS/MP region Tag addition with File.

Command

- AtmMPTool FILE (OTP|NVDS) <file> TAG SET_CAP <value> <new file>

Parameter	
<value>	0~31
<new file>	File path to save.
Result	<new file> will be saved and shows success.

9.30 Set Testing Log Refresh

The production test will write detailed logs in the log folder. Use this command to create a new log file of the current time and import subsequent logs into the file.

Command
• AtmMPTool LOG REFRESH

Parameter	
Result	the next log will be written to the new log file.

9.31 Set Testing Log Finished

Call this command to update the final status on the file name of the log

Command
• AtmMPTool LOG FINISHED

Parameter	
Result	update log file name if this testing has an error message.

9.32 CLI detail log

The MP Tool CLI mode displays simple test results in the standard output. If you need a more complete test LOG, you can find the “(DevID)_CLI_<test_time>.xml” file in logs/xml/ under the same directory of AtmMPTool.exe, which contains a more detailed log.

You can use Set Testing Log Refresh and Set Testing Log Finished to cut the log into different files according to the test flow.

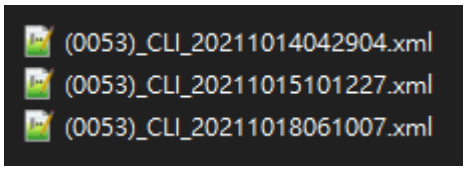


Figure 9-19 CLI Command Detail Logs

9.33 Write MP region

Call this command to update the final status on the file name of the log

Command
• AtmMPTool APT <DevID> DUT FLASH MPREGION <mp_region_file> <mp_region_addr> <mp_region_max_size> <mp_region_lock_size>

Parameter	
<mp_region_file>	bin file to be programmed.
<mp_region_addr>	start address for mp region
<mp_region_max_size>	max size for mp region
<mp_region_lock_size>	protection size for mp region
Result	The binary content of the MP region file will be written to the specified address.

***Note**: The BD address tag in MP region would take effect during firmware initialization period.*

9.34 Pull MP region

Call this command to pull the binary of the MP region from flash in the IC.

Command
• AtmMPTool APT <DevID> DUT FLASH PMPREGION <mp_region_file> <mp_region_addr> <mp_region_max_size>

Parameter	
<mp_region_file>	bin file to be programmed.
<mp_region_addr>	start address for mp region
<mp_region_max_size>	max size for mp region
Result	Save the binary content of the MP region in the IC's flash to the file.

9.35 Set UART baud rate setting

Call this command to set the baud rate to the CLI common setting, this setting will continue until the next time this command is called again.

Command
• AtmMPTool SETTING BAUDRATE <baud rate>

Parameter	
<baud rate>	UART baud rate
Result	Update global setting.

9.36 Dump flash to bin file

Dump flash to a bin file.

Command
• AtmMPTool APT <DevID> DUT FLASH DUMP <bin file> (<addr>) (<size>)

Parameter	
<bin file>	output file name
<addr>	start address (default is 0x0)
<size>	size (default is dump to end)
Result	Save the binary content of the IC flash to a file.

9.37 Flash chip erase

Erase all contents of flash.

Command
• AtmMPTool APT <DevID> DUT FLASH CHIP_ERASE

Parameter	
Result	Erase all contents of flash.

9.38 Check protecting

Check flash is protecting or not

Command
• APT <DevID> DUT FLASH CHECK_PROTECTING

Parameter	
Result	If it fails, show error status and its error reason.

9.39 Show ATM information

Show the information from atm file

Command
• AtmMPTool FILE ATM <file> SHOW

Parameter	
<file>	Input atm file path
Result	Display the information on the console.

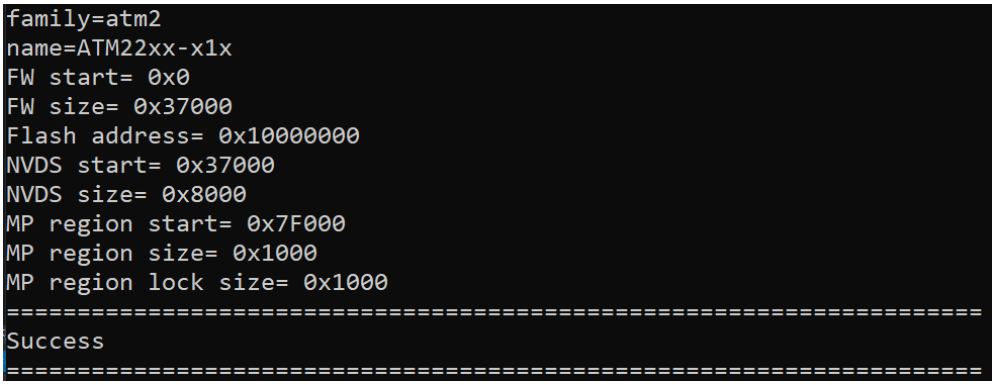


Figure 9-20 Information from ATM File

9.40 Export FW bin file from ATM file

Export FW bin file from ATM file

Command
• AtmMPTool FILE ATM <in_file> EXP_FW <out_file>

Parameter	
<in_file>	input atm file path
<out_file>	output bin file path
Result	Export FW bin to file path.

9.41 Export NVDS bin file from ATM file

Export NVDS bin file from ATM file

Command
• AtmMPTool FILE ATM <in_file> EXP_NVDS <out_file>

Parameter	
<in_file>	input atm file path
<out_file>	output bin file path
Result	Export NVDS bin to file path.

9.42 FTDI pin control

Command
• AtmMPTool APT <DevID> AG FTDI_CONTROL (NO_POWER) (<CONTROL_FLOW...>)

Parameter	
NO_POWER	If set this parameter, default VBAT, VCCIO, LS pin would set to low
<CONTROL_FLOW…>	Follow the format <ACTION_LABLE> <delay_ms> <ACTION_LABLE> <delay_ms> <ACTION_LABLE>.... to control these pins. The valid ACTION_LABLE: • VBAT_H : control VBAT pin to high

	• VBAT_L : control VBAT pin to low • VCCIO_H : control VCCIO pin to high • VCCIO_L : control VCCIO pin to low • LS_H : control LS pin to high • LS_L : control LS pin to low • BBOOT_H : control BBOOT pin to high • BBOOT_L : control BBOOT pin to low • PWD_H : control PWD pin to high • PWD_L : control PWD pin to low Example 1 - sends AtmMPTool APT 50 AG FTDI_CONTROL NO_POWER VBAT_H 50 VCCIO_H 150 LS_H to: 1. Default VBAT, VCCIO, LS are low 2. Set VBAT to high 3. delay 50 ms 4. Set VCCIO to high 5. delay 150 ms 6. Set LS to high Example 2 sends AtmMPTool APT 50 AG FTDI_CONTROL BBOOT_H 20 PWD_H 0 PWD_L 20 BBOOT_L to: 1. Default VBAT, VCCIO, LS are high 2. Set BBOOT to high 3. delay 20 ms 4. Set PWD to high 5. delay 0 ms 6. Set PWD to low 7. delay 20 ms 8. Set BBOOT to low
Result	Export NVDS bin to file path.

9.43 DUT RRAM Firmware Programming

Program firmware to RRAM.

Command
• AtmMPTool APT <DevID> DUT RRAM FW <mpbin file>

Parameter	
FW	FW - Program firmware and verification.
<mpbin file>	File pathname. (.mpbin), E.g, C:\\AtmosicMP\\Test.mpbin for firmware. The MP bin file context is: —---------- [files] app=<elf or bin file path of app> app_start=<start address of app, if use elf please set 0>

	bootloader=<elf or bin file path of bootloader> bootloader_start=<start address of bootloader, if use elf please set 0> ——---------- You can refer the file "C:\AtmosicMP\fw\MP_33xx-5.mpbin"
Result	No error message

9.44 DUT RRAM NVDS Programming

Program NVDS data to RRAM.

Command
• AtmMPTool APT <DevID> DUT RRAM NVDS <file> <addr> • AtmMPTool APT <DevID> DUT RRAM NVDS <file>

Parameter	
<file>	NVDS file pathname. (.bin). E.g, C:\\AtmosicMP\\flash_nvds.bin.
<addr> Optional	NVDS start address. This information is customer specific. If not provided, the default address of NVDS is 0x78000.
Result	No error message

10. Multiple DUT Testing in CLI Mode

The Atmosic CLI command can execute testing/burning a maximum of eight devices at the same time.

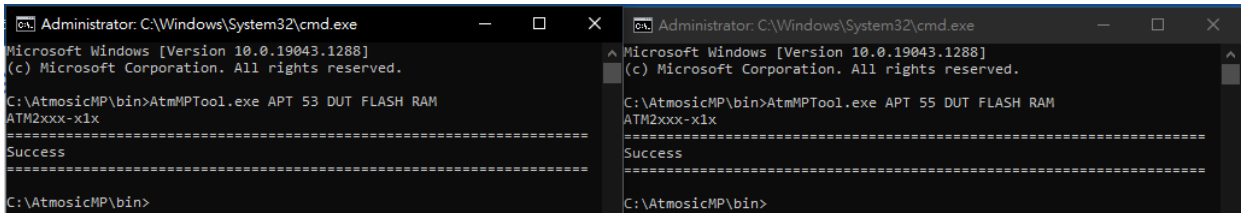


Figure 10-1 Multiple DUT Testing Under CLI Mode¶

11. Power Control

11.1 Using the MPTool GUI

The GUI power control is automatic. The GUI options and configuration are similar to previous versions.

11.2 Using the MPTool console mode

11.2.1 Using Download Board

The download board “DL_V5”(or later), and the MPTooll version v2.0.14 (or later) supports DUT’s power control. It is important to update the following 3 processes:

- On application opening

  Run this command before setting the DUT on the fixture

  Set all FTDI pins to low (same as power-off)

  CMD: `AtmMpTool.exe APT <DEVID> AG FTDI_CONTROL LS_L 450 VCCIO_L 50 VBAT_L`

- Before testing

  Change FTDI pins to trigger DUT power-up

  CMD: `AtmMpTool.exe APT <DEVID> AG FTDI_CONTROL VBAT_H 50 VCCIO_H 450 LS_H`

- After testing

  Set all FTDI pins to low

  CMD: `AtmMpTool.exe APT <DEVID> AG FTDI_CONTROL LS_L 450 VCCIO_L 50 VBAT_L`

For more information about the FTDI pins control command, refer to FTDI pin control section.

11.2.2 Using APT Board

When using the APT board, the power control is controlled by the pins of the AG.

- On application opening

  Run this command before setting the DUT on the fixture

  Set all FTDI pins to low (same as power-off)

CMD:

- AtmMPTool.exe APT %DevID% AG RESET
- AtmMPTool.exe APT %DevID% AG IO SETUP 17 1
- AtmMPTool.exe APT %DevID% AG IO SETUP 18 1
- AtmMPTool.exe APT %DevID% AG IO SET 17 0
- AtmMPTool.exe APT %DevID% AG IO SET 18 0

- Before testing

  Change FTDI pins to trigger DUT power-up

  CMD:

  - AtmMPTool.exe APT %DevID% AG IO SET 17 1
  - AtmMPTool.exe APT %DevID% AG IO SET 18 1

- After testing

  Set all FTDI pins to low

  CMD:

  - AtmMPTool.exe APT %DevID% AG IO SET 17 0
  - AtmMPTool.exe APT %DevID% AG IO SET 18 0

12. NVDS Tool

Apart from the FILE command in AtmMptool.exe, NVDS tool provides another way to modify NVDS tags and generate the files which are used for Flash NVDS / OTP or MP region. The executable file is located in “C:\AtmosicMP\bin\nvds_tool.exe”.

Users may use this tool for three types of operations. See Table 12-1.

Table 12-1 nvds_tool Options List

Display tags of NVDS or OTP file	
-r <file>	Add from file. For OTP, the “-i” should be added in front of “-r”.
-v	Verbose output (with values of tags and data)
-p	Pretty output (with readable meaning of tags and data)

-b	Binary output
-i	Inverted. Used for OTP file, e.g., “nvds_tool -i -t otp_nvds.nvm”
Add, erase or lock tags	
-t <tag>	Add tag, e.g., “nvds_tool -t 01 -d addr.bin” for adding BDADDR tag.
-d <file>	Add data associated with the tag. This option should follow the -t tag.
-e <tag>	Erase tag.
-l <tag>	Lock tag. Once a tag is locked in file , it cannot be unlocked through this tool.
Generation new NVDS or OTP file	
-o <file>	Output to file. Only allowed with “-b” output options.

12.1 DISPLAY Content of NVDS and OTP file

There are four NVDS/OTP file output display formats.

If no output option applied, the default output would be ASCII hex digitals serially. Figure 12-1 shows the NVDS and OTP file ASCII output:



Figure 12-1 NVDS Display with ASCII Output

With the binary “-b” option, the output would be binary data serially. Figure 12-2 shows the NVDS and OTP file binary output:

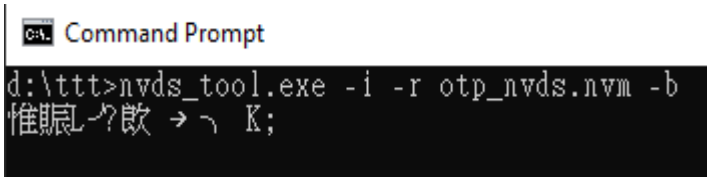


Figure 12-2 NVDS Display with Binary Output

Binary output is also used to save a new file. Please refer to the section Generate new OTP/NVDS/MP region file for more information.

***Note**: The garbled characters in Figure 12-2 can be ignored as they are caused by the NVDS Tool forcing the use of ASCII code to display the byte array.*

With the verbose ”-v” option, the output would be the ASCII HEX value of each tag and data with address offset. Figure 12-3 shows the NVDS and OTP file verbose output:

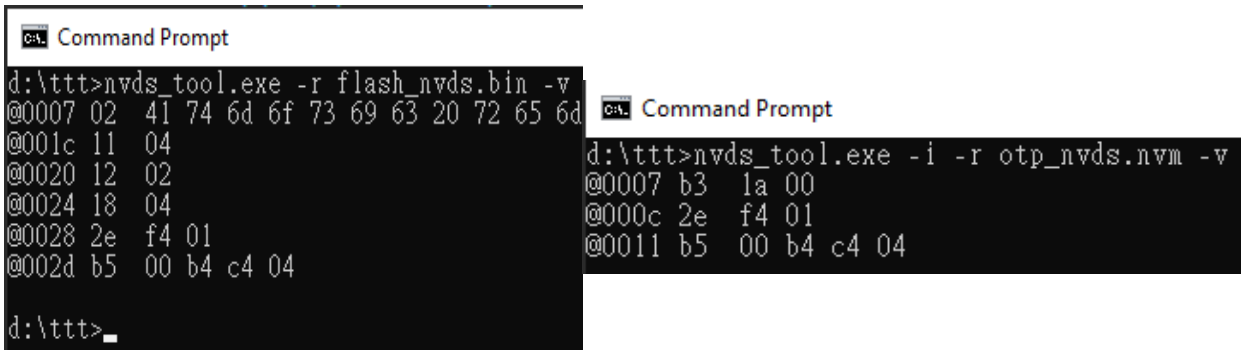


Figure 12-3 NVDS Display with Verbose Output

With the option “-p” , the output would be the readable string of each tag and data with address offset. Figure 12-4 shows the NVDS and OTP file pretty output:

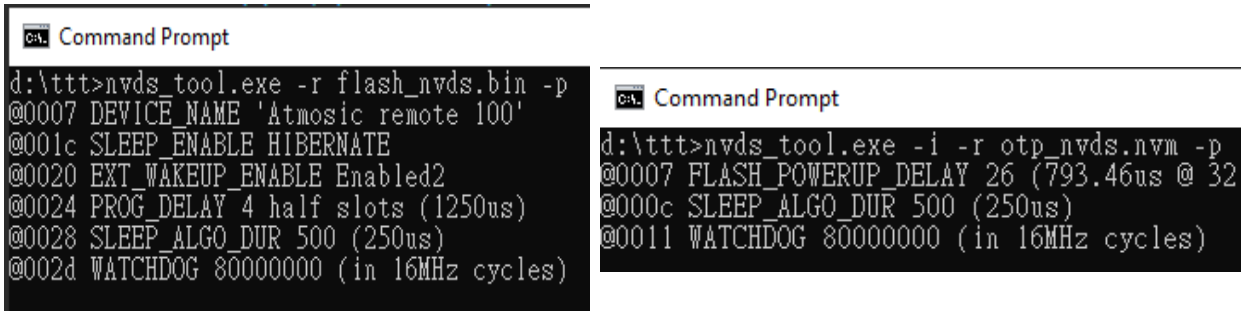


Figure 12-4 NVDS Display with Pretty Output

12.2 Add, Erase Tag

The add tag option “-t <tag> -d <data file >” is used to replace or add tag. If the tag exists, the old tag will be erased and a new tag will be created at the end.

The BD_ADDRESS (0x01) tag is an example to demonstrate how to assign a new BT address 11:22:33:44:55:66 through this option.

1) Apply “-i -r otp_nvds.nvm -p” to show the otp_nvds.nvm OTP file. The output contains BD_ADDRESS with 7C:50:43:3E:65:9A content. See Figure 12-5.

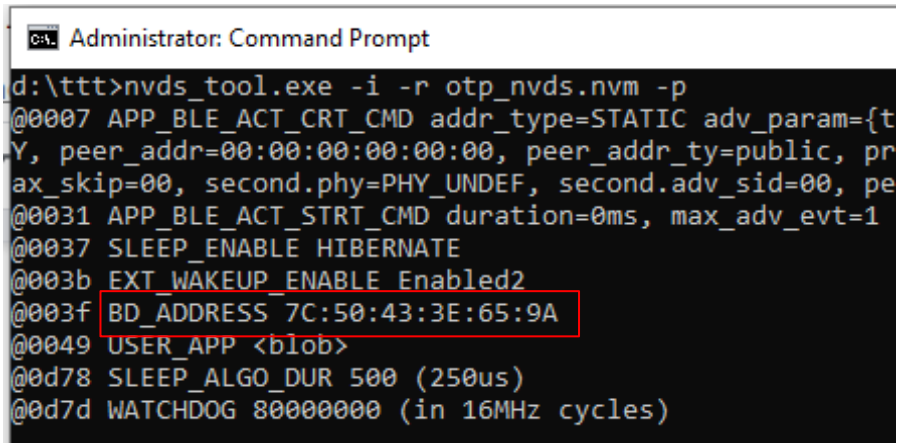


Figure 12-5 otp_nvds.nvm Content

2) Prepare a binary file for data of the new BD_ADDRESS tag. See Figure 12-6.

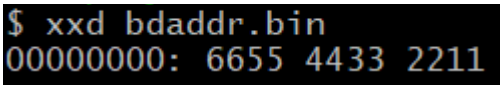


Figure 12-6 Binary File with New BD_ADDRESS Tag

3) Add “-t 01 -d bdaddr.bin” on the command line and see the address changed to 11:22:33:44:55:66. See Figure 12-7.

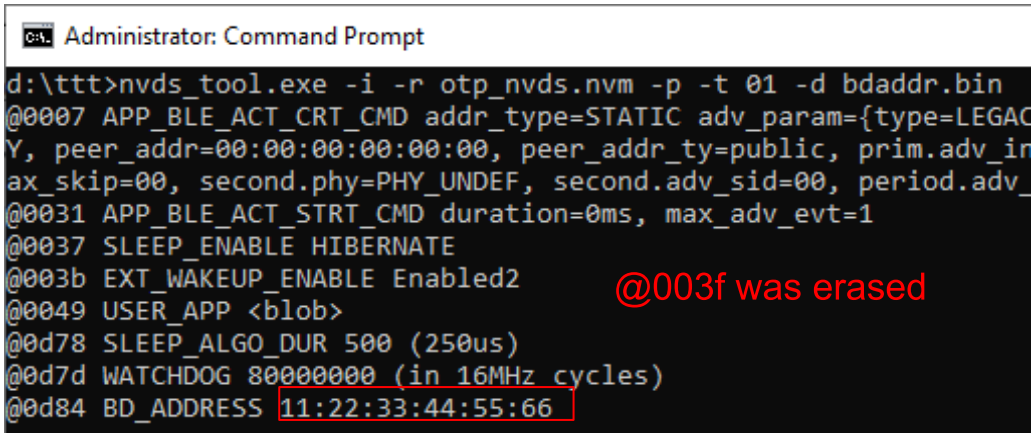


Figure 12-7 BD_ADDRESS Change

The erase tag operation marks the tag invalid. If the command line of the add tag example is modified from “-t 01 -d bdaddr.bin” to “-e 01”, the BD_ADDRESS tag will not show on the output. See Figure 12-8.

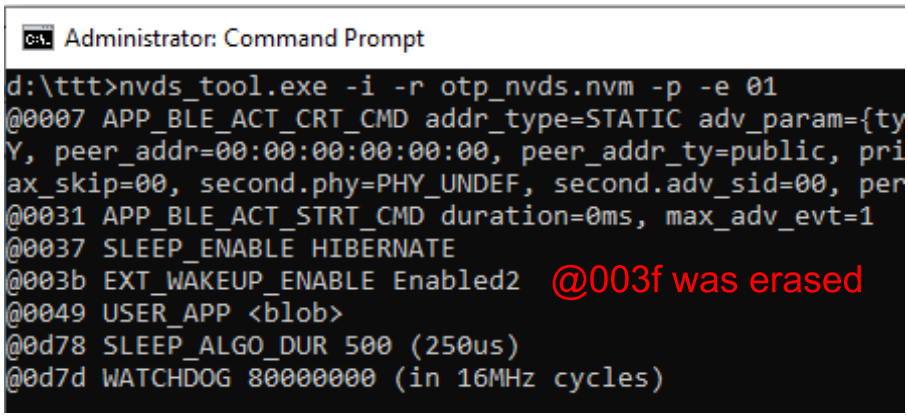


Figure 12-8 Erase BD_ADDRESS (01) Tag

For more information on tag ID, please refer to section Common OTP/NVDS/MP Region Tag ID.

12.3 Generate New OTP/NVDS/MP region File

The “-o <file>” option is used for generating new OTP or NVDS/MP region files. It should be used with the binary output “-b” option. In the previous add tag example, If change the -p to “-b -o otp_nvds_new.nvm”, then a new file called otp_nvds_new.nvm will be generated. See Figure 12-9.

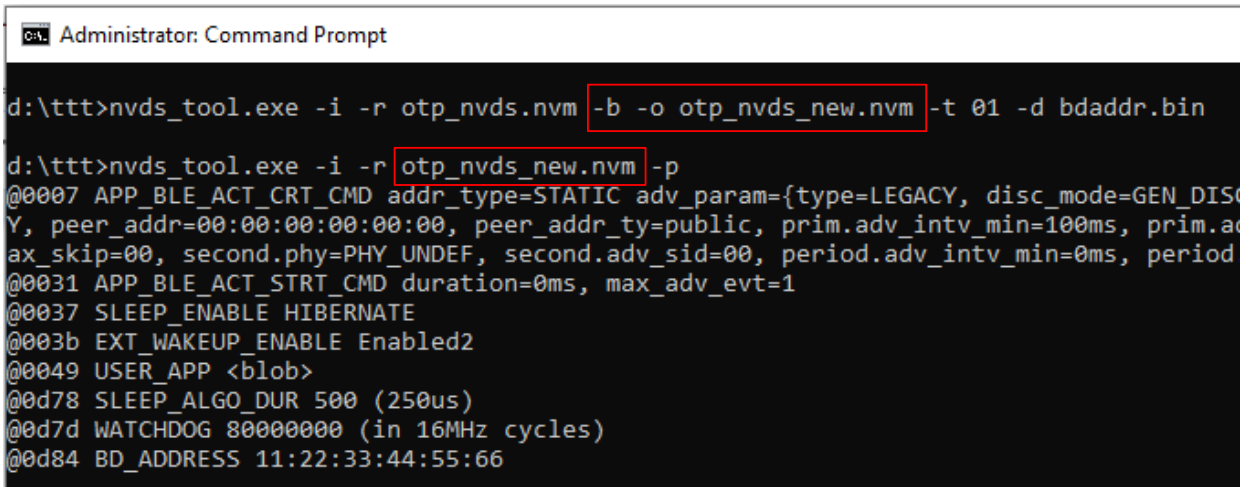


Figure 12-9 Generate a New OTP File

13. Common OTP/NVDS/MP Region Tag ID

In sections OTP/NVDS Tag addition with File and Generate New OTP and NVDS/MP region File , Tag ID can refer to Table 12-2.

Table 12-2 NVDS/MP Region and OTP Tag ID

NVDS / MP Region and OTP	
BD address	0x01
NVDS	
Device name	0x02
MP Region	
CAP	0xAE
OTP	
CAP	In OTP data, CAP value is a part of Tag 0xBD and is not independent bytes, it is recommended to use Set 16M Crystal CAP to modify the CAP value.

14. Troubleshooting

1) Figure 14-1 shows error log “No board found”

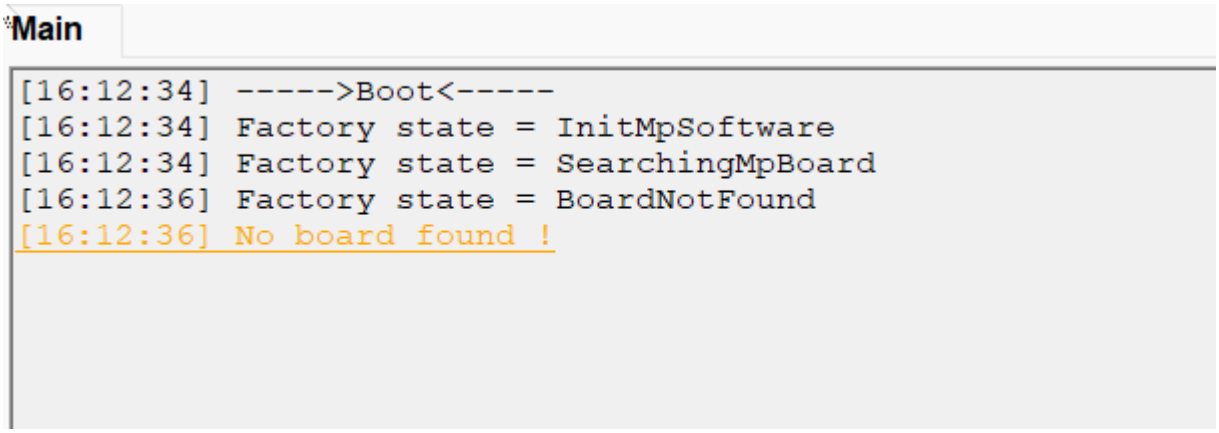


Figure 14-1 No Board Found

Please make sure the usb cable is connected. And, open the device manager to check if “libusbK USB Devices” exists. If not, please refer to section Install MFx Interface to install USB driver.

2) Figure 14-2 shows error log “Different type of boards found simultaneously ”.

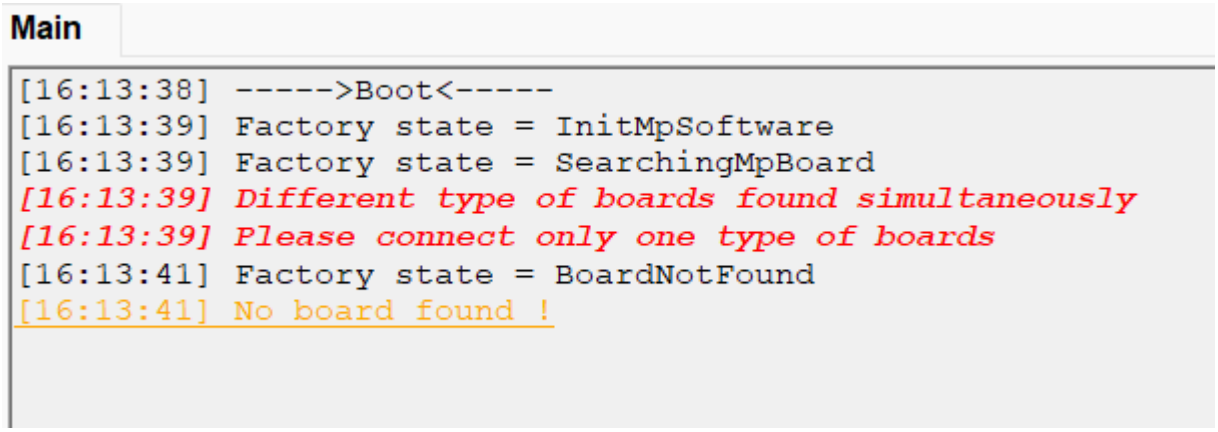


Figure 14-2 Different Type of Boards Found Simultaneously Error Log

Please plug in with only one type of board and click the refresh button again.

3) Figure 14-3 shows error (< 100) on Burn_Ram

Test Item	DUT(0016)
Burn_Ram	fail (46)
Crystal Trim	
FOS_1M	
TX_1M	

Figure 14-3 Burn_Ram Error

Please refer to section Install MFx Interface to install USB drivers.

4) Figure 14-4 shows “Invalid board <num> found” and the num is bigger than 32768.

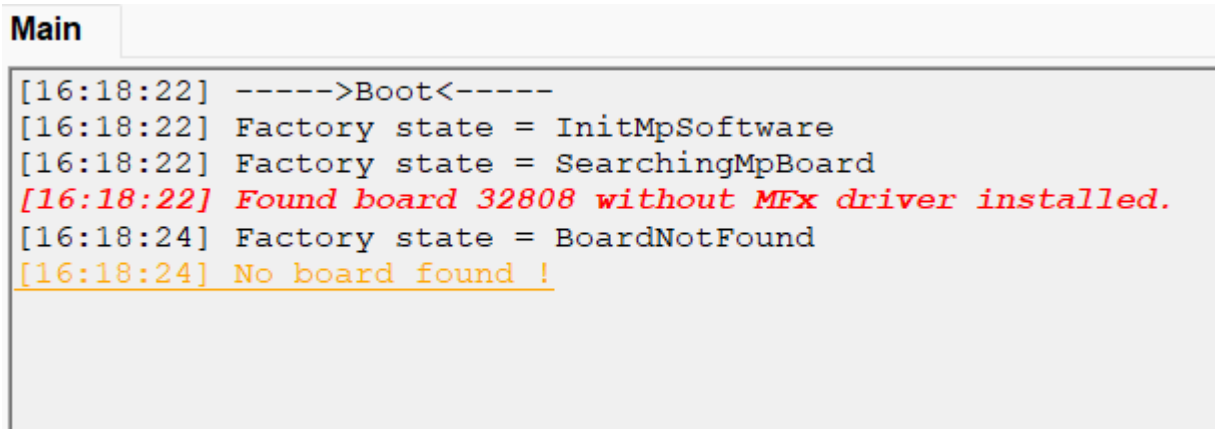


Figure 14-4 Invalid Board <num> Found

Please refer to the section Install MFx Interface to install USB drivers.

Revision History

Date	Version	Description
August 8, 2022	0.57	Added DL Board Versions, Uninstall any previous versions of the MPTool, GUI Mode Quick Start, FTDI pin control, DUT RRAM Firmware, Programming sections. Changed format, corrected typos.
December 14, 2021	0.56	Format change. Added Dump flash to bin file, Flash chip erase, Check protecting, Show ATM information, Export FW bin file from ATM file, Export NVDS bin file from ATM file sections.
November 10, 2021	0.55	Updated section NVDS/MP Region Create Blank File. Added sections System Environment Variable, Pull MP region and Set UART baud rate setting.
October 25, 2021	0.54	Updated Write MP Region, Setting Tab, Figure 19 - Run Tab Before Testing, Figure 20 - Run Tab Under Testing, Figure 21- Run Tab After Testing Success, Figure 22 - Run Tab After Testing Fail, Figure 23 - Setting Tab, Figure 24 - Crystal Trim Setting, Figure 34 - User Firmware/Tags Setting, Figure 39 - Start Option, added Figure 41 - MP Region Information, Figure 42 - Write Calibration Data to MP Region, Figure 43 - Choose Which Calibration Data to Write, Figure 64 - CLI Command Detail Logs, Set Testing Log Refresh, Set Testing Log Finished, Write MP region, Multiple DUT Testing in CLI Mode, Common OTP/NVDS/MP Region Tag ID, updated NVDS Tool.
September 30, 2021	0.53	Updated Figure 8 - Start Menu Shortcuts, Figure 19 - Run Tab Before Testing
July 23, 2021	0.52	Format change; updated Setting Menu, Figure 14 - MP Tool Setting, Help Menu, Figure 18 - MP Tool Help Menu, Figure 20 - Setting Tab, Figure 34 - User Firmware/Tags Setting, User Firmware (.elf)/NVDS file (.bin)/Tags Customization; added sections Environment Setup for DL, Adjust Serial Port Latency Timer, UART0 Pin Modification, Start Option, Atmosic Production Tool Console Mode, NVDS Tool
May 11, 2020	0.51	Added sections OTP file (.nvm), Troubleshooting.

March 25, 2020	0.50	Initial version created.

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